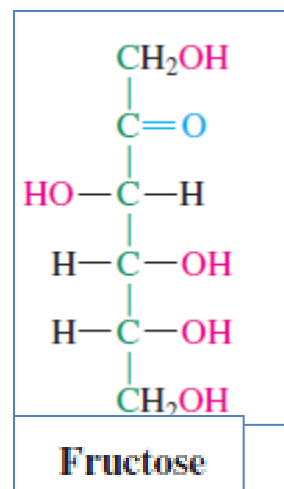
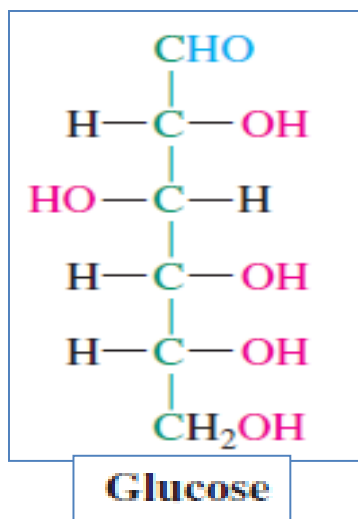


Chapter 2: Carbohydrates

- **Contents**
 - 2.1. Introduction
 - 2.2. Classification of carbohydrates
 - 2.3. Monosaccharides
 - 2.3.1. Introduction (classification and nomenclature)
 - 2.3.2. D and L Designations of monosaccharides
 - 2.3.3. Cyclic structures of monosaccharides
 - 2.3.4. Mutarotation
 - 2.3.5. Reactions of monosaccharides
 - 2.4. Disaccharides
 - 2.3.1. Glycosides
 - 2.3.2. Sucrose
 - 2.3.3. Lactose, Maltose and Cellobiose
 - 2.5. Oligosaccharide
 - 2.6. Polysaccharides
 - 2.6.1. Classification of polysaccharides
 - 2.6.2. Nomenclature of polysaccharides
 - 2.6.3. Starch
 - 2.6.4. Cellulose

Carbohydrates

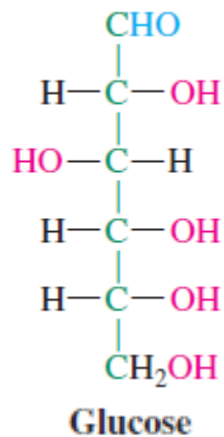
- ❑ General molecular formula $\text{C}_n\text{H}_{2n}\text{O}_n = \text{C}_n(\text{H}_2\text{O})_n$
- ❑ Appeared to be *hydrates of carbon*
- ❑ Later structural studies indicated that they do not contain molecules of water



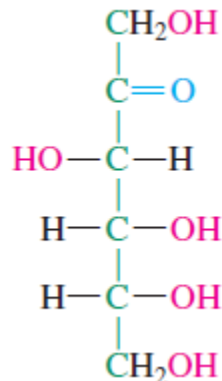
• Carbohydrates:

– *polyhydroxy of*

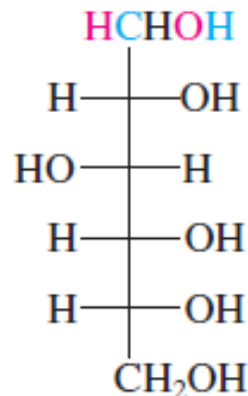
- *aldehydes,*
- *ketones or acids or*
- *their derivatives,*
- *together with linear and cyclic polyhydroxyls*



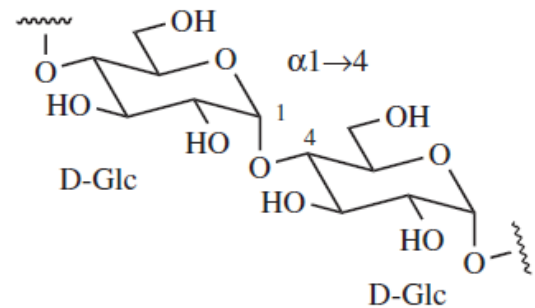
Glucose



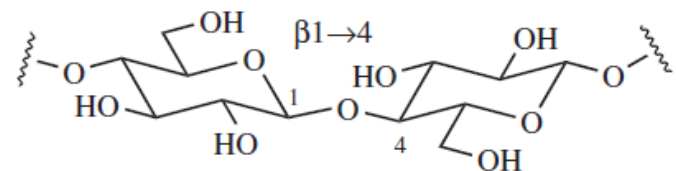
Fructose



D-Glucitol
(D-Sorbitol)

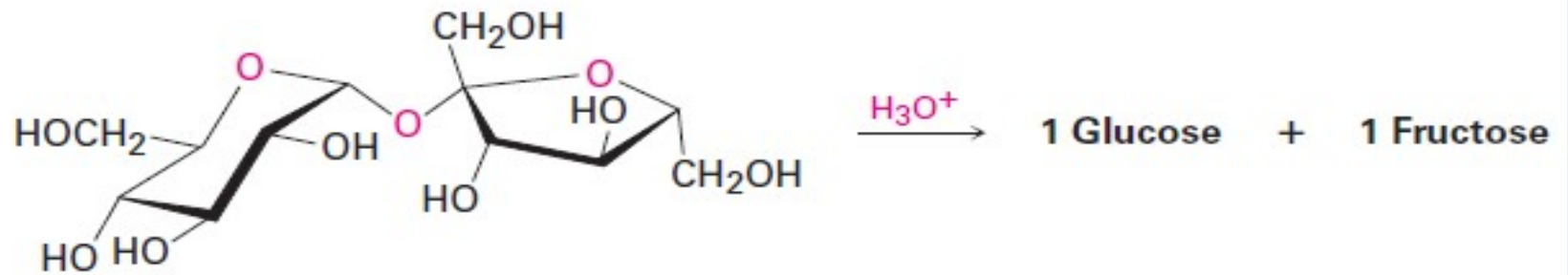


amylose
(1000–2000 residues)

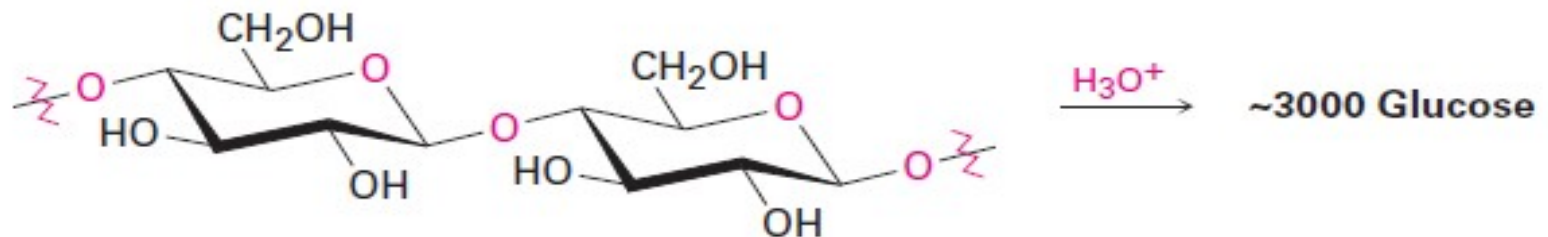


cellulose
(~8000 residues)

- substances that **hydrolyze** to yield polyhydroxy aldehydes and ketones.



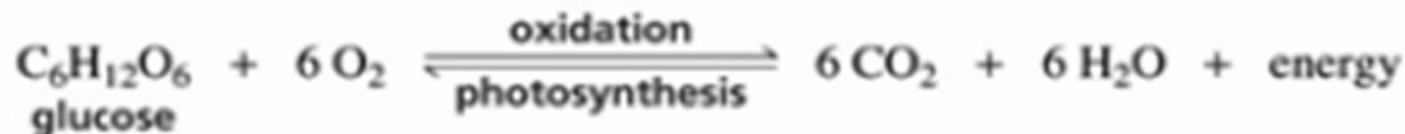
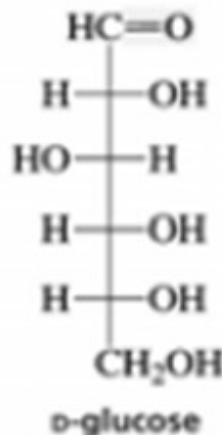
Sucrose
(a disaccharide)



Cellulose
(a polysaccharide)

D-Glucose in Nature

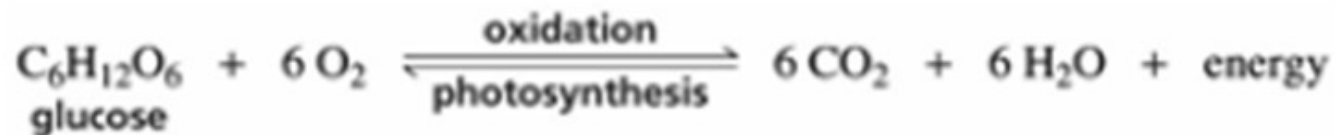
- The most abundant carbohydrate is D-glucose
- Cells of organisms oxidize glucose for energy



- In animals excess glucose is converted to a polymer called glycogen

D-Glucose Sources

- Plants convert excess glucose into starch
- Animals get glucose from eating plants
- Plants obtain glucose by making it in a process known as photosynthesis

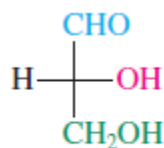


2.2. Classification of carbohydrates

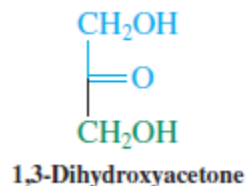
- Based on **chemical structure and degree of polymerization**
 - Monosaccharides and their derivatives,
 - Disaccharides (2 monosaccharide units),
 - Oligosaccharides (3-9 monosaccharide units), and
 - Polysaccharides (more than 10 monosaccharide units)

Monosaccharides

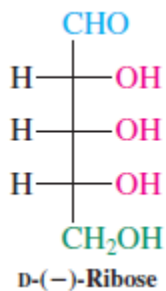
- a carbohydrate that cannot be hydrolyzed to a simpler carbohydrate
- they have the general formula $C_nH_{2n}O_n$, where n varies from 3 to 8
- chiral **polyhydroxyalkanals** or **polyhydroxyalkanones** (except 1,3-di hydroxyacetone)



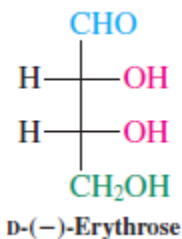
D-(+)-Glyceraldehyde



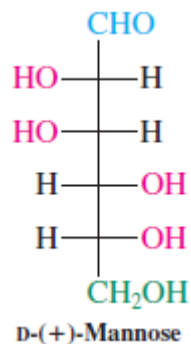
1,3-Dihydroxyacetone



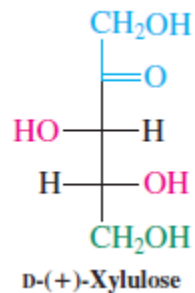
D-(-)-Ribose



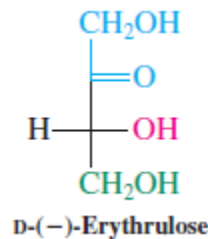
D-(-)-Erythrose



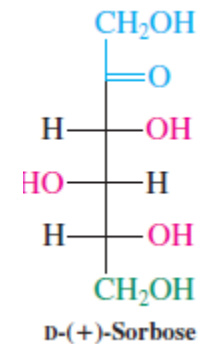
D-(+)-Mannose



D-(+)-Xylulose



D-(-)-Erythrulose

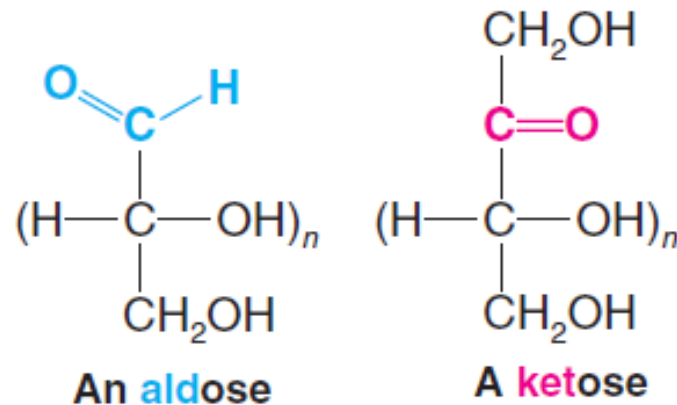


D-(+)-Sorbose

- **Classified based on:**

- **Functional groups**

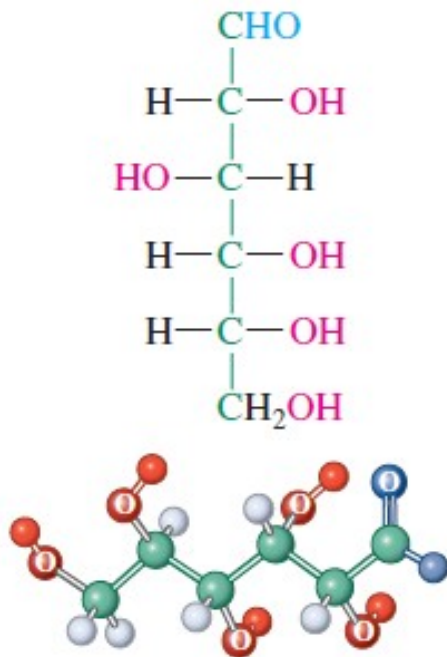
- ✓ aldose: a monosaccharide containing an aldehyde group
 - ✓ ketose: a monosaccharide containing a ketone group



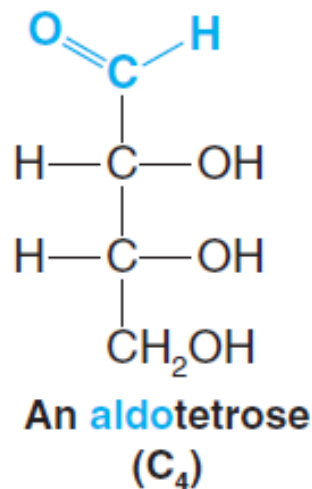
- **Number of carbons**

- trioses (three carbons), tetroses (four carbons), pentoses (five carbons), hexoses (six carbons), etc

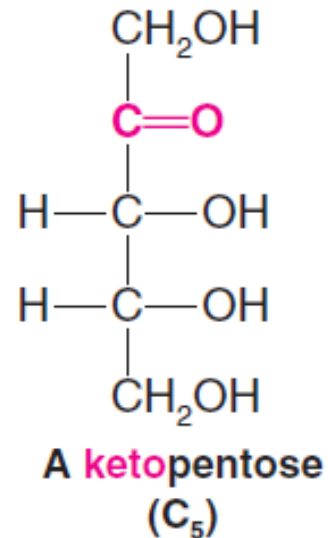
- Combination of the two is usually used:



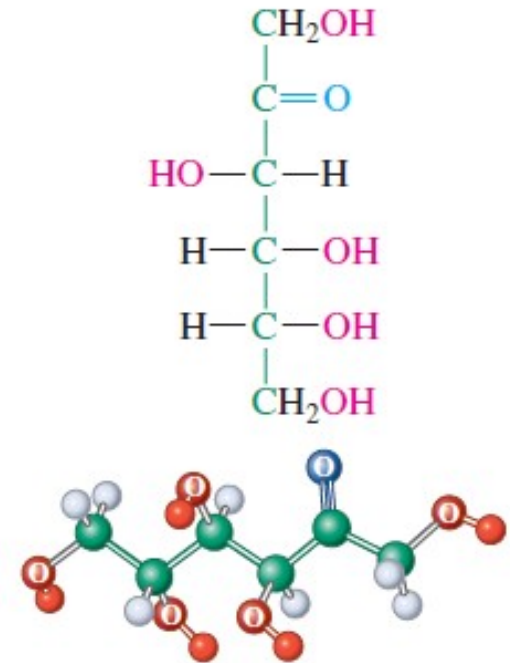
Glucose
(An aldohexose)



An aldotetrose
(C₄)



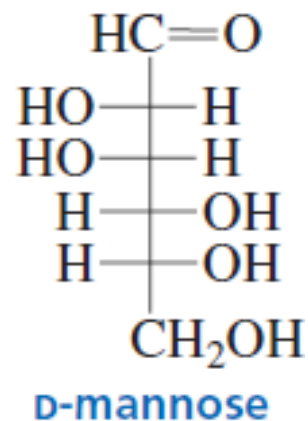
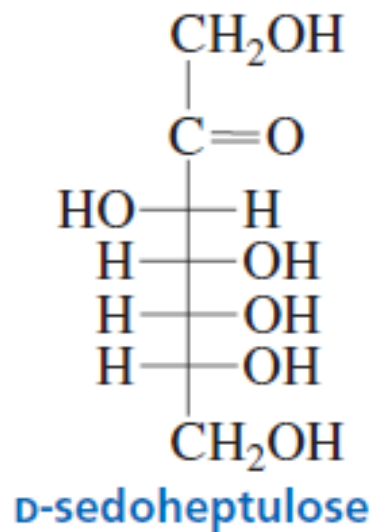
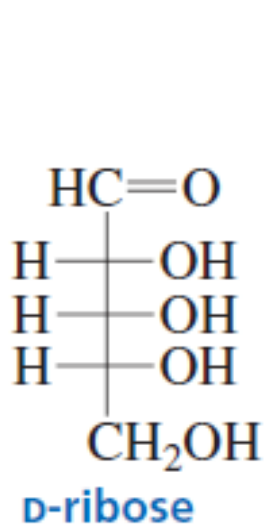
A ketopentose
(C₅)



Fructose
(A ketohexose)

- **Exercise**

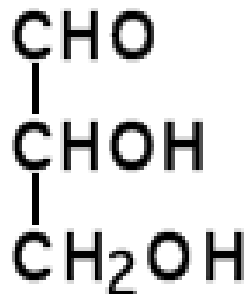
- Classify the following monosaccharides:



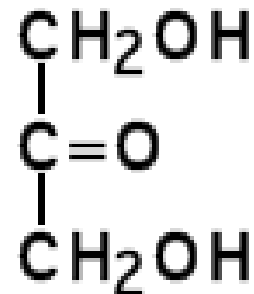
2.3.2. D and L Designations of monosaccharides

Monosaccharides

- There are only two trioses



Glyceraldehyde
(an aldotriose)

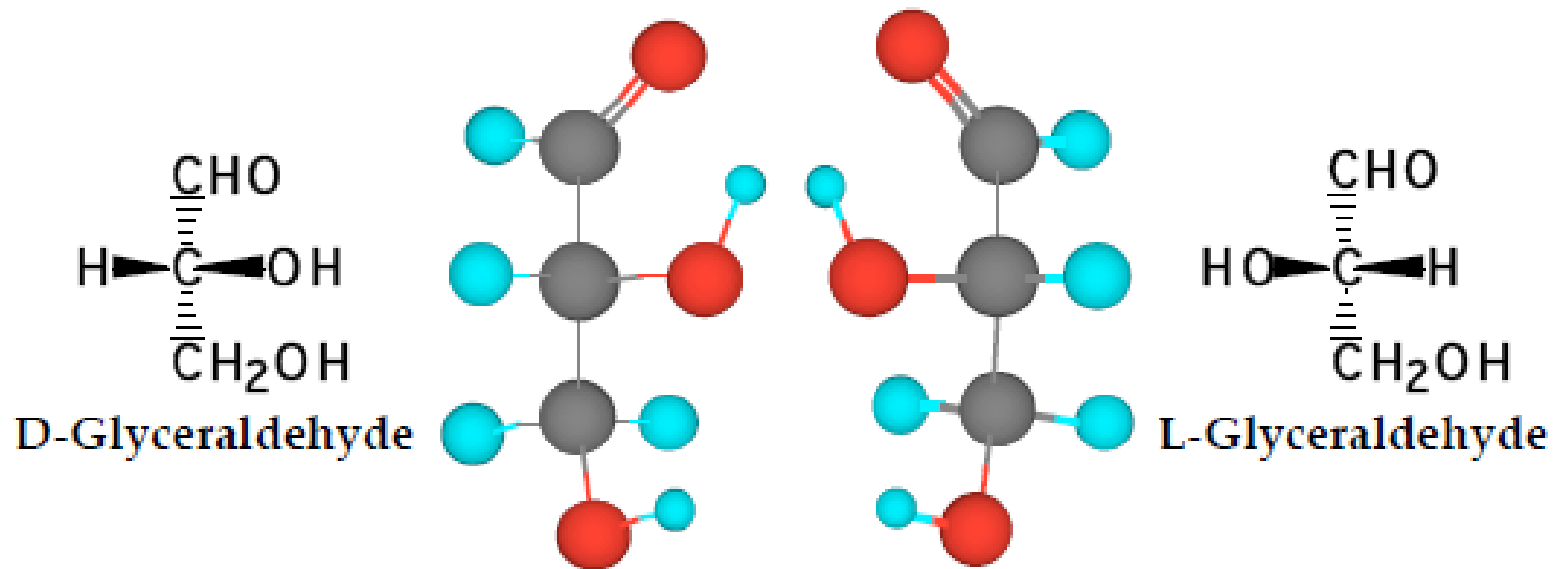


Dihydroxyacetone
(a ketotriose)

- Simplest monosaccharides

Monosaccharides

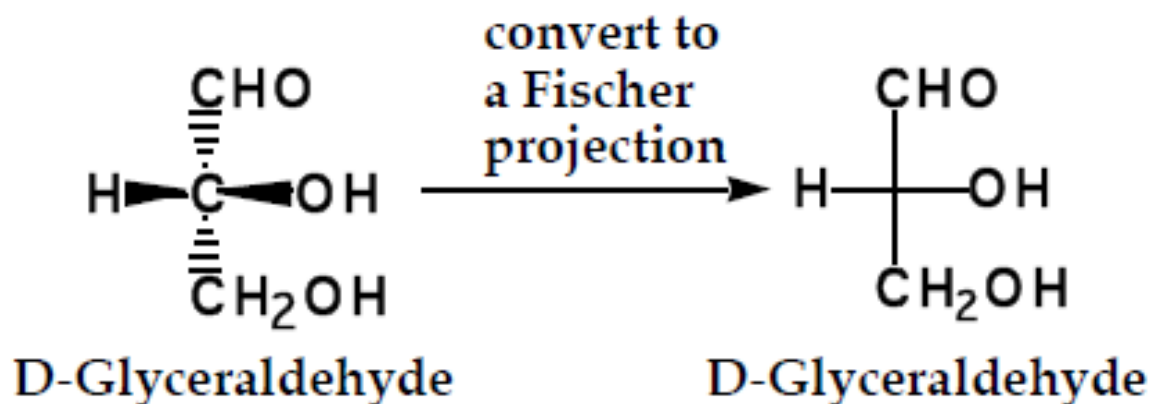
- Glyceraldehyde contains a stereocenter and exists as a pair of enantiomers



✓ $2^1 = 2$ enantiomers

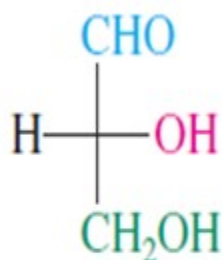
Fischer Projections

- **Fischer projection:** a two dimensional representation for showing the configuration of tetrahedral stereocenters
 - horizontal lines represent bonds projecting forward
 - vertical lines represent bonds projecting to the rear
 - the carbon atom at the intersection of the horizontal and vertical lines is not shown

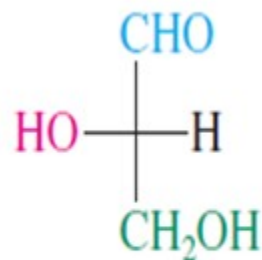


D,L Monosaccharides

- According to the conventions proposed by Fischer
 - **D-monosaccharide**: a monosaccharide that, when written as a Fischer projection, has the -OH on its penultimate carbon on the right
 - **L-monosaccharide**: a monosaccharide that, when written as a Fischer projection, has the -OH on its penultimate carbon on the left

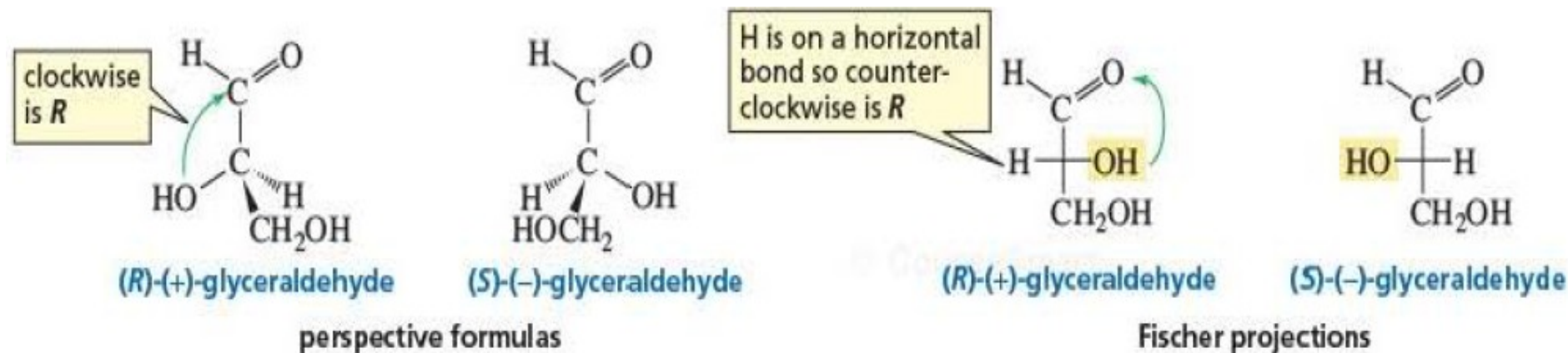
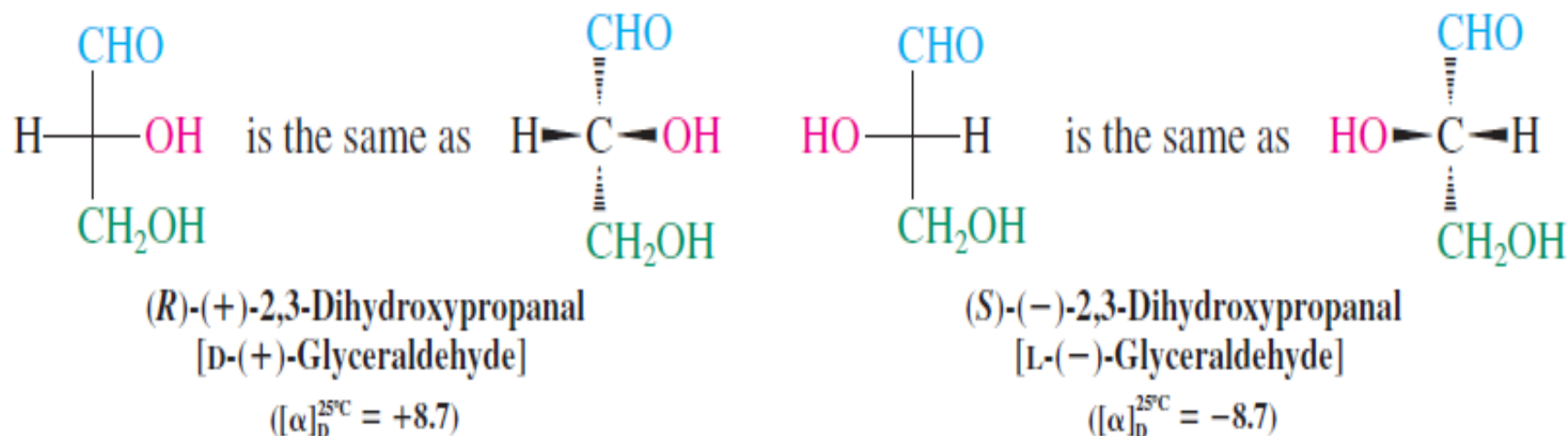


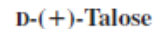
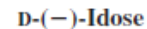
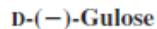
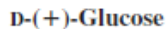
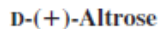
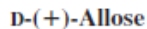
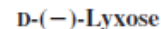
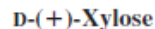
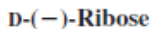
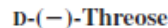
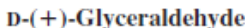
D-(+)-Glyceraldehyde



L-(-)-Glyceraldehyde

Fischer Projections of the Two Enantiomers of 2,3-Dihydroxypropanal (Glyceraldehyde)



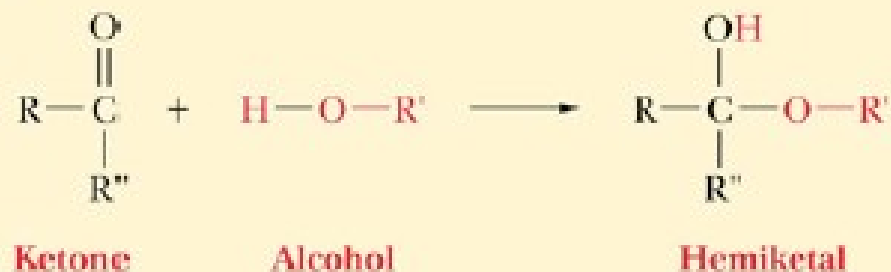


2.3.3. Cyclic structures of monosaccharides

Cyclic Structure

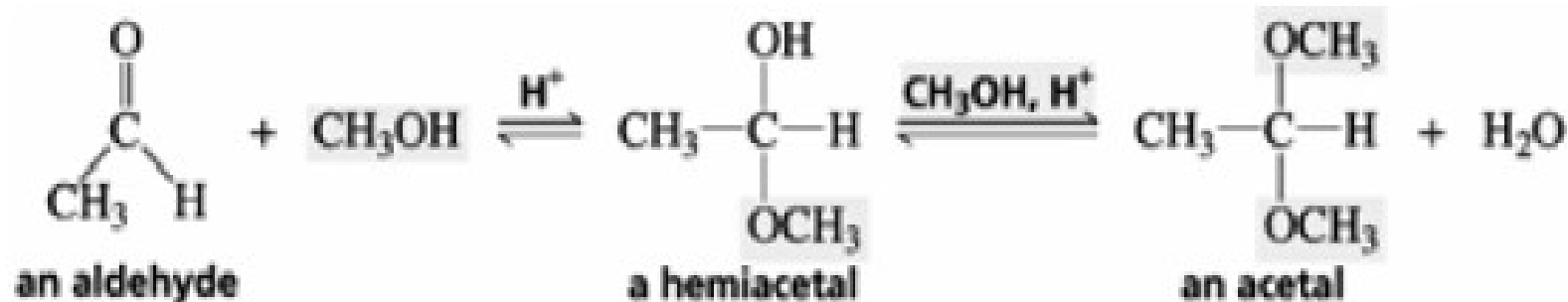
- Monosaccharides have -OH and C=O groups in the same molecule and exist almost entirely as five- and six-membered cyclic hemiacetals
 - **anomeric carbon**: the new stereocenter resulting from cyclic hemiacetal formation
 - **anomers**: carbohydrates that differ in configuration only at their anomeric carbons

General reaction

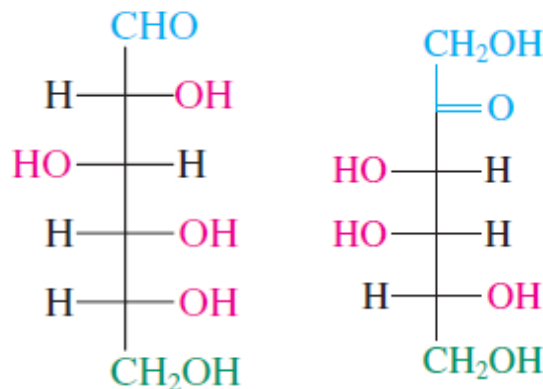


Cyclic Structures/Hemiacetal Formation

- Recall that any aldehyde group can react with an alcohol to form a hemiacetal

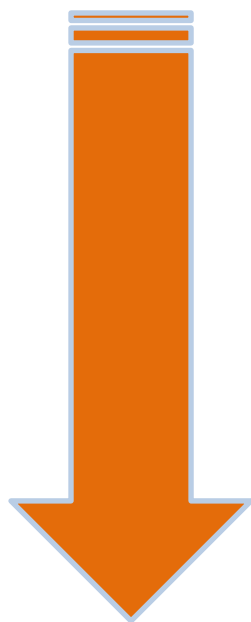


Multifunctional groups

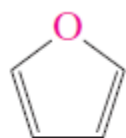


Intramolecular reaction between active groups

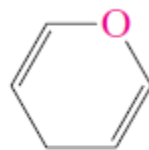
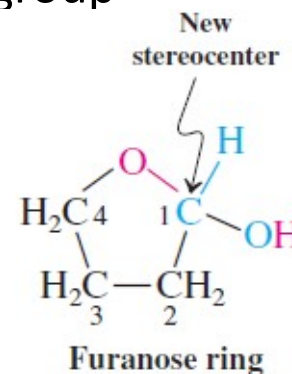
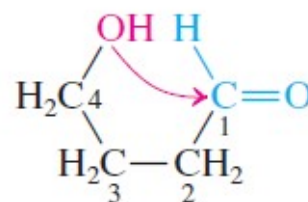
- ☞ nucleophilic addition reaction between a -OH and a C=O group
- ☞ hemiacetals and hemiketals



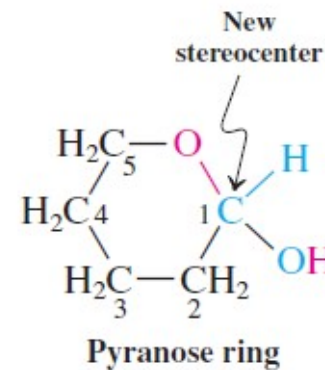
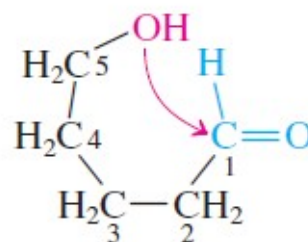
Cyclic structures



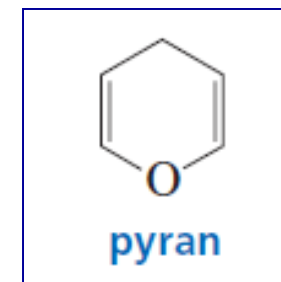
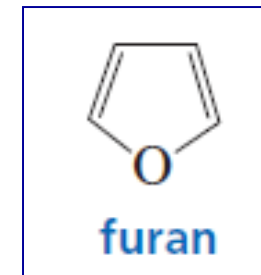
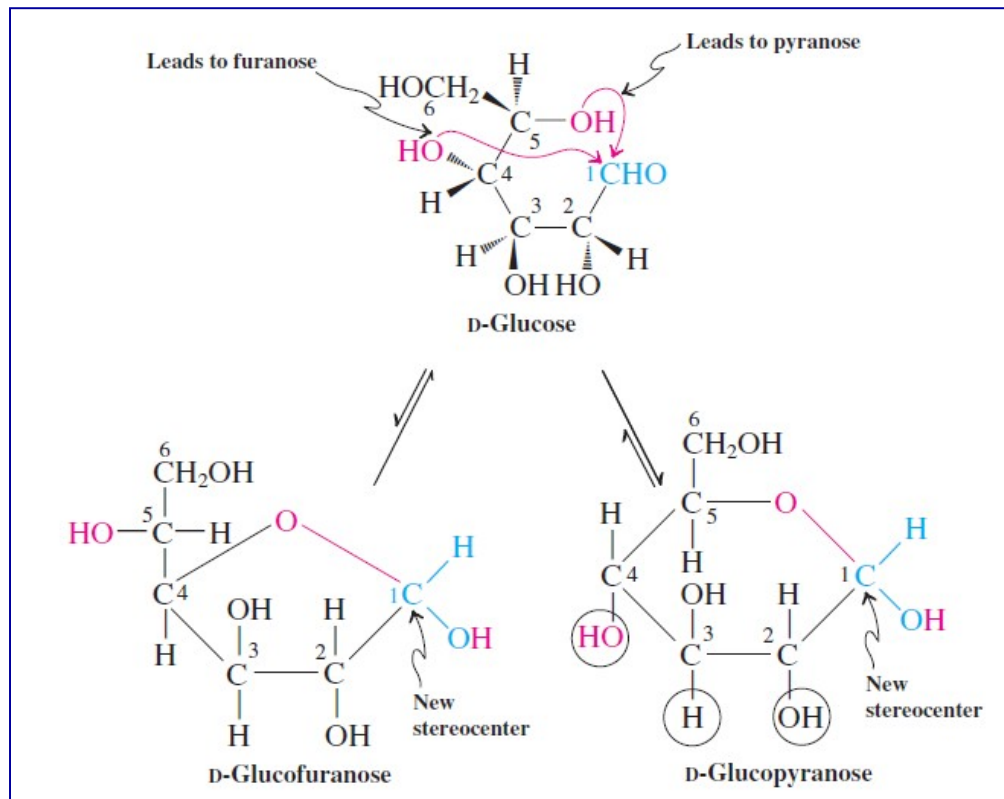
Furan



Pyran



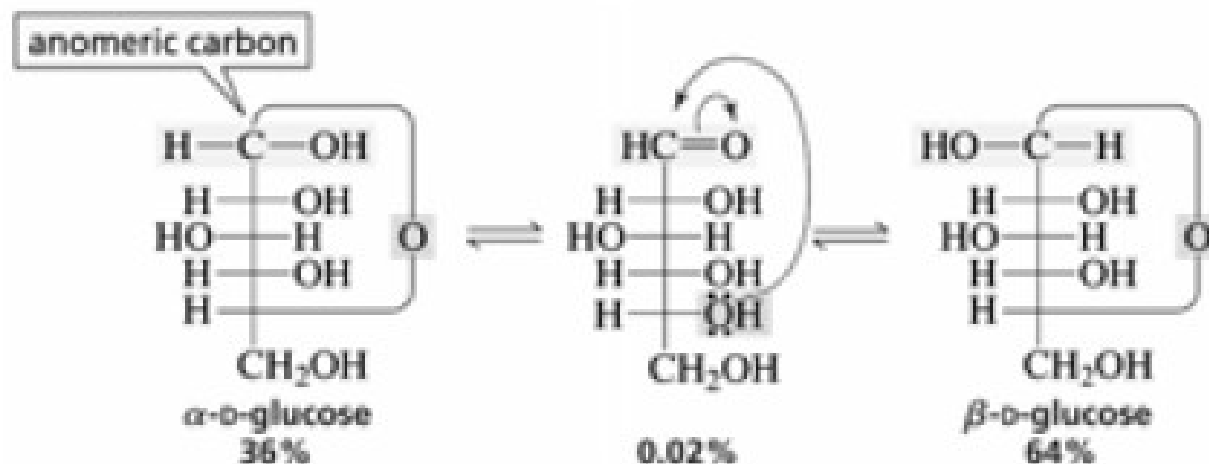
- In fact in solution exist as cyclic forms
 - Six-membered $\Rightarrow$ Pyranose
 - Five membered $\Rightarrow$ Furanose



- **Carbonyl group carbon**
 - formation becomes chiral center due to hemiacetal or ketal
 - **Anomers**
 - Diastereomers that differ at one stereocenter
 - Example: anomers of glucose (next slide)

Cyclic Structures/Hemiacetal Formation

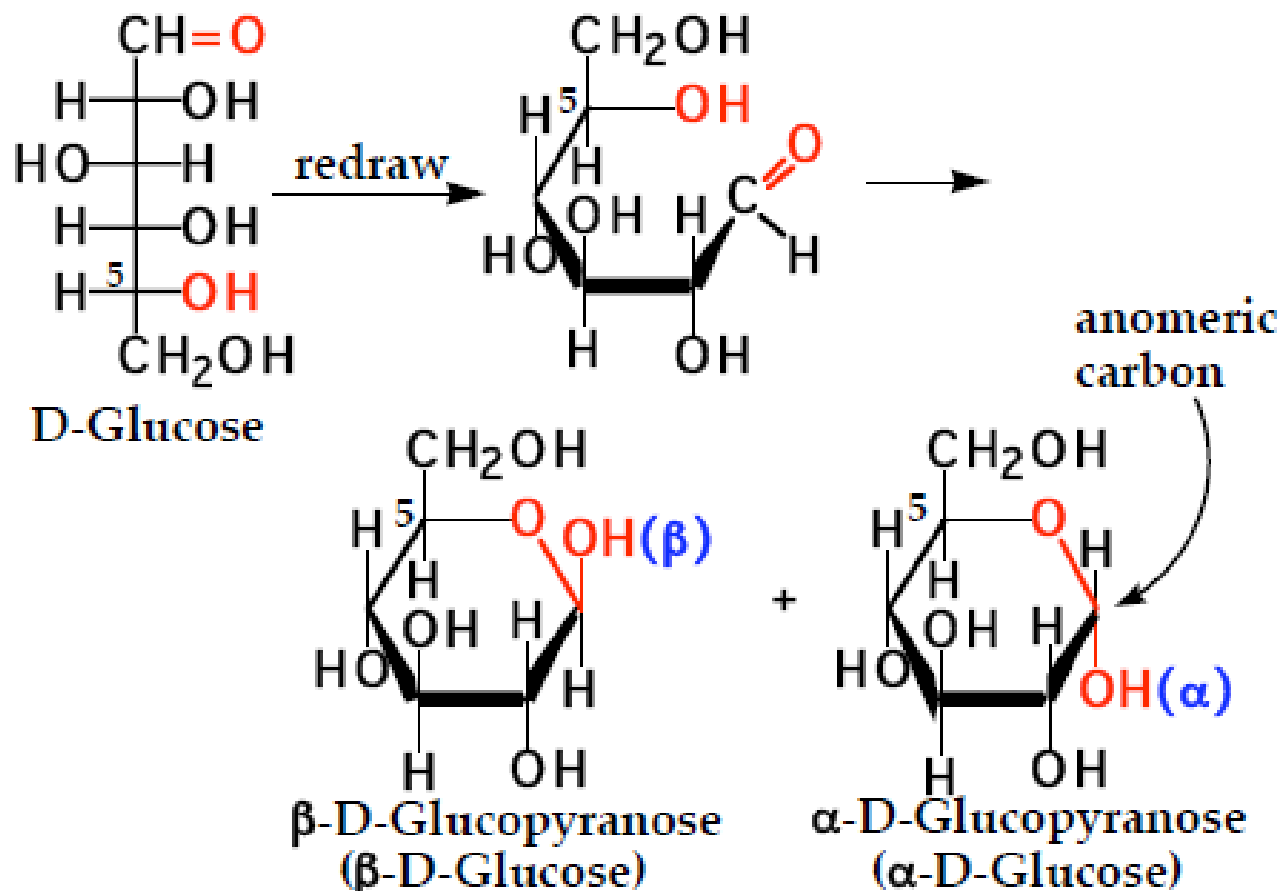
- A hydroxyl at C-5 reacts with the carbonyl, forming a hemiacetal

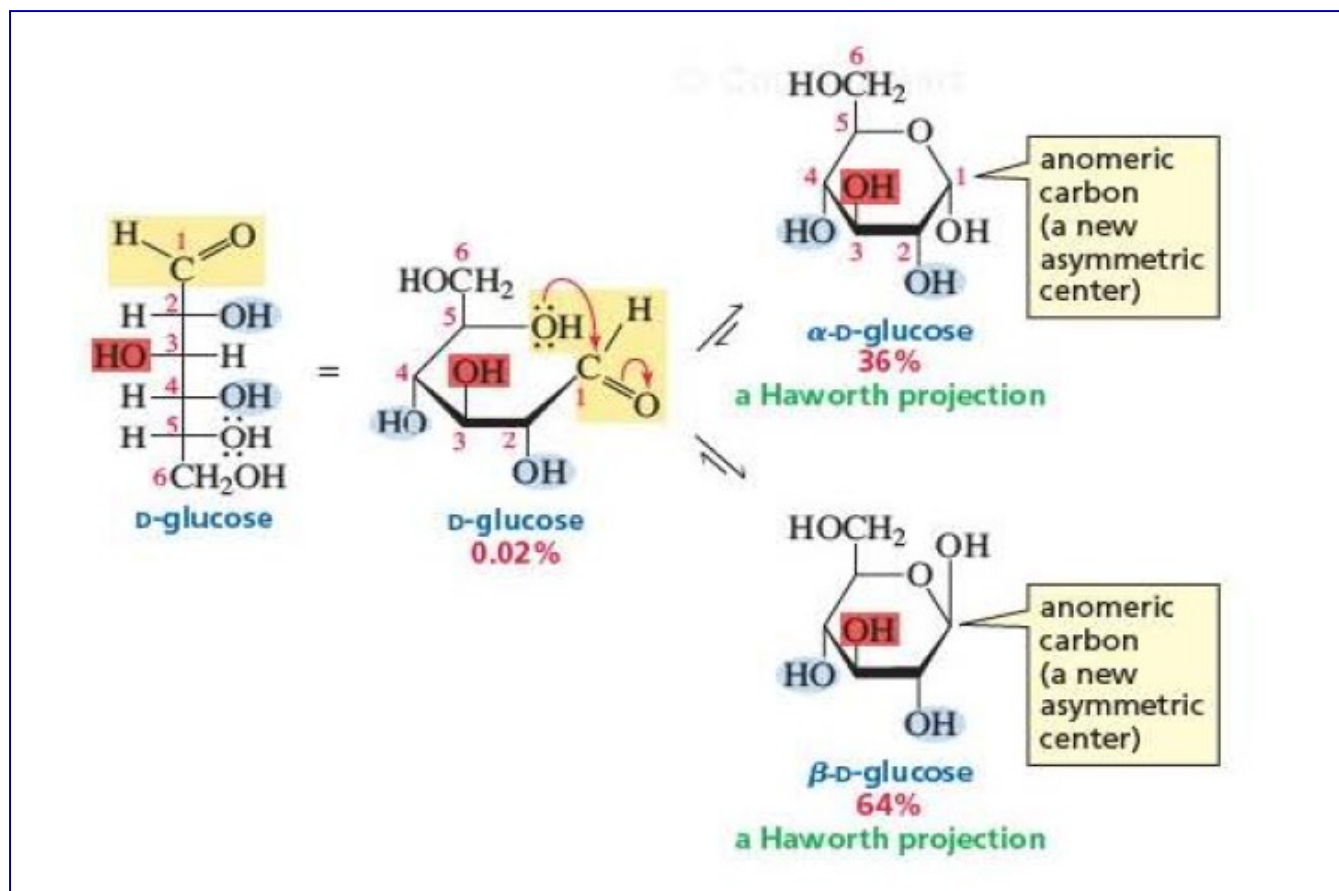
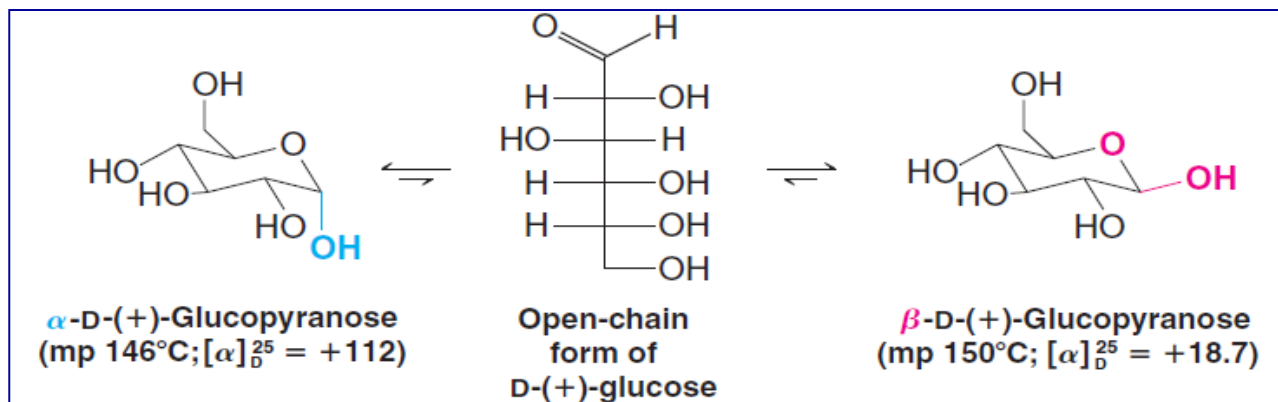


Haworth Projections

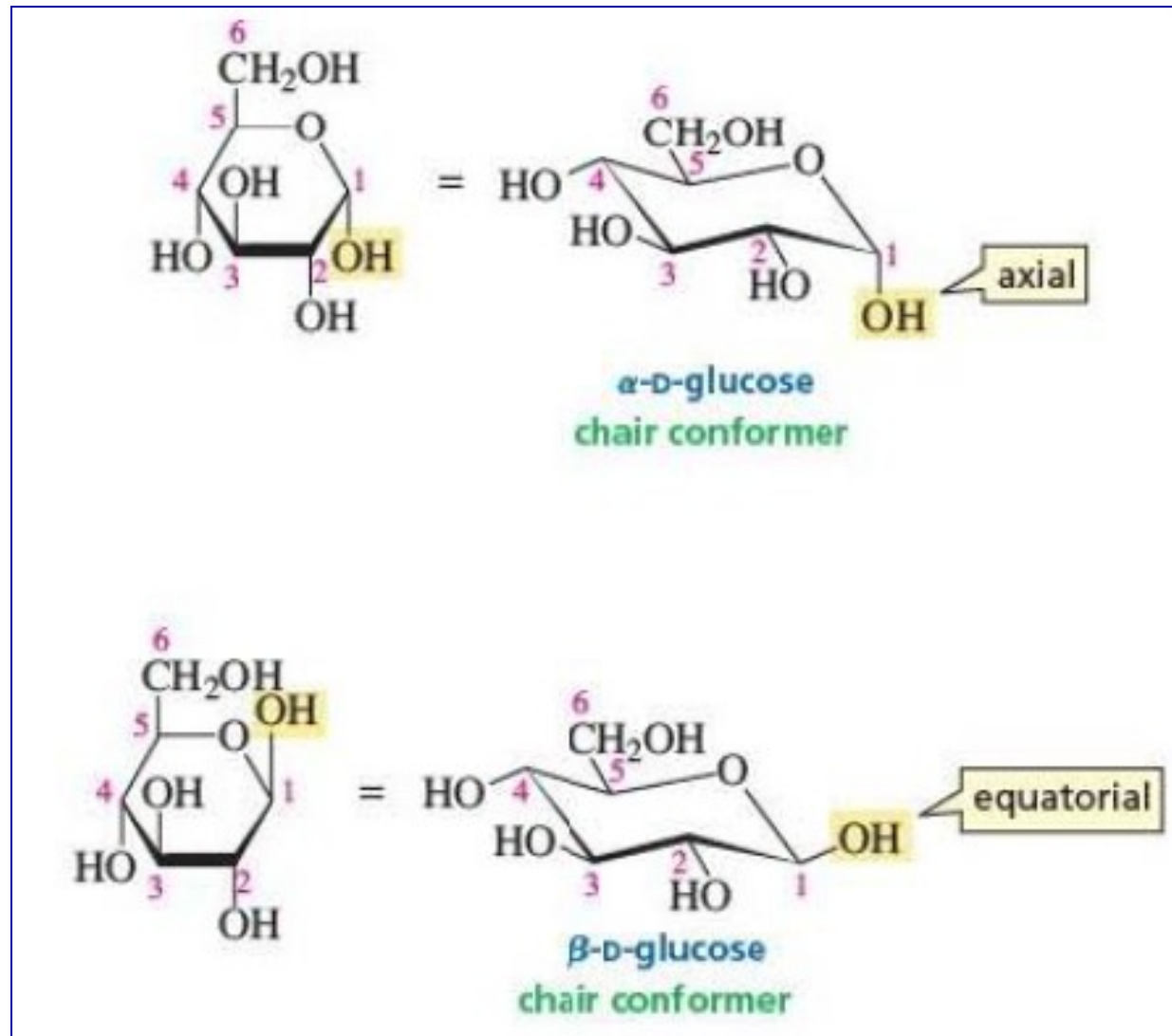
- **Haworth projections**
 - five- and six-membered hemiacetals are represented as planar pentagons or hexagons, as the case may be, viewed through the edge
 - most commonly written with the anomeric carbon on the right and the hemiacetal oxygen to the back right
 - the designation β - means that -OH on the anomeric carbon is cis to the terminal -CH₂OH; α - means that it is trans

Haworth Projections

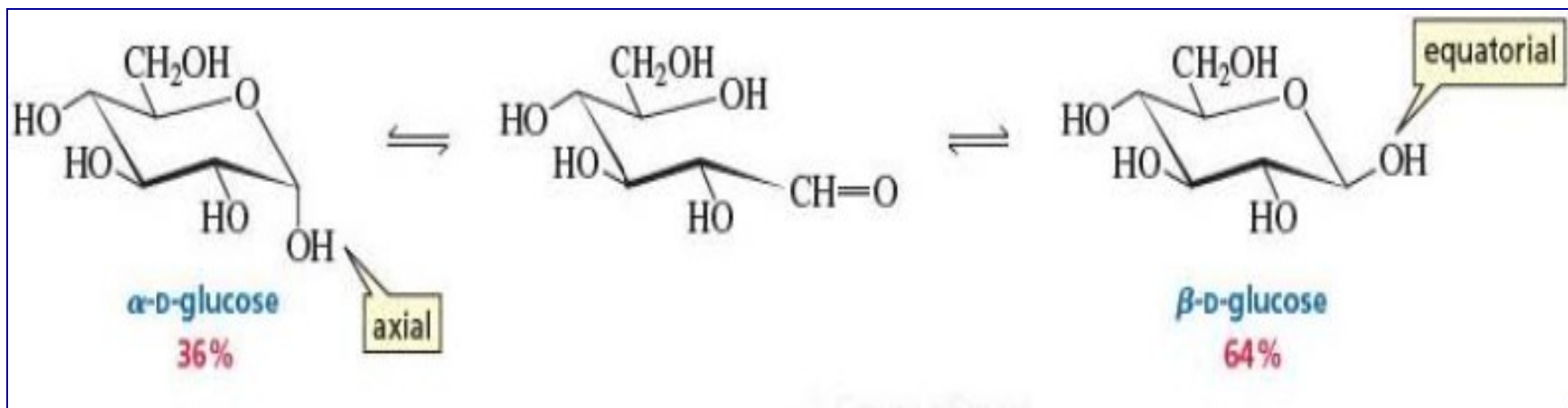




- Chair conformation



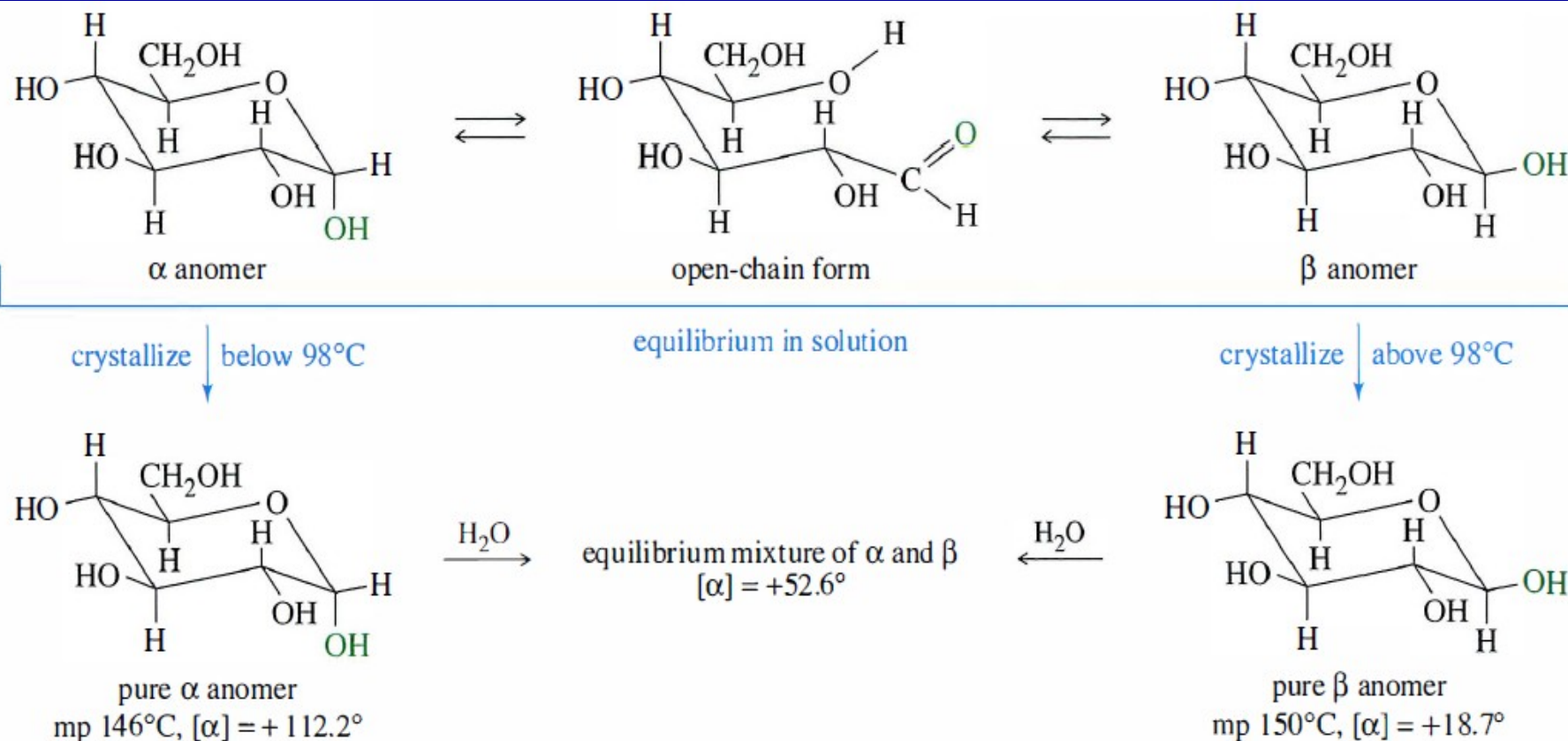
- *Why is the β anomer of glucopyrananose in higher amount than α anomer?*



2.3.4. *Mutarotation*

• Mutarotation

- A slow change in optical rotation to an equilibrium value



- **Exercise: Calculating specific rotations of diastereomeric sugar mixtures**
- **Question:**
 - Calculate how much of the α ($[\alpha] = + 112.2^\circ$) anomer and how much of the β ($[\alpha] = + 18.7^\circ$) anomer are present in an equilibrium mixture with a specific rotation of $+ 52.7^\circ$.

- **Solution:**

- Note:

- The specific rotation of an equilibrium mixture is the average of the rotations of the contributing isomers — however, *weighted* by the respective mole fractions in the mixture.

- Let:

- x_α % of α anomer ($[\alpha] = + 112.2^\circ$) and
 - x_β % of β anomer ($[\alpha] = + 18.7^\circ$)

- Then

- $x_\alpha + x_\beta = 100\%$ or $x_\beta = 100 - x_\alpha$

– Then:

$$+ 52.7 = (+112)(x_{\alpha}) + (+18.7)(x_{\beta})$$

– Substituting $100 - x_{\alpha}$ *for* x_{β}

$$+ 52.7 = (+112)(x_{\alpha}) + (+18.7)(100 - x_{\alpha})$$

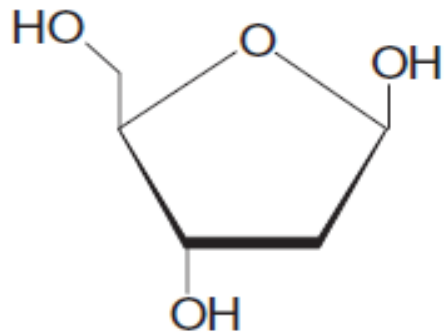
– Solving for x_{α}

- $x_{\alpha} = 36\%$ *and* $x_{\beta} = 64\%$

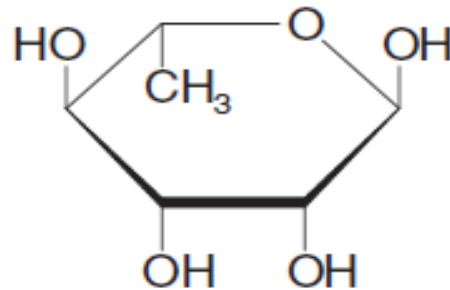
2.3.5. *Amino and deoxy sugars*

- **Deoxy sugars**

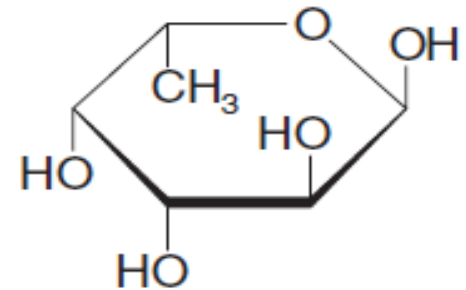
- Deoxy sugar is sugar in which a hydroxyl group is replaced by a hydrogen
- Methylene group or a methyl group



β -2-Deoxy-D-ribose

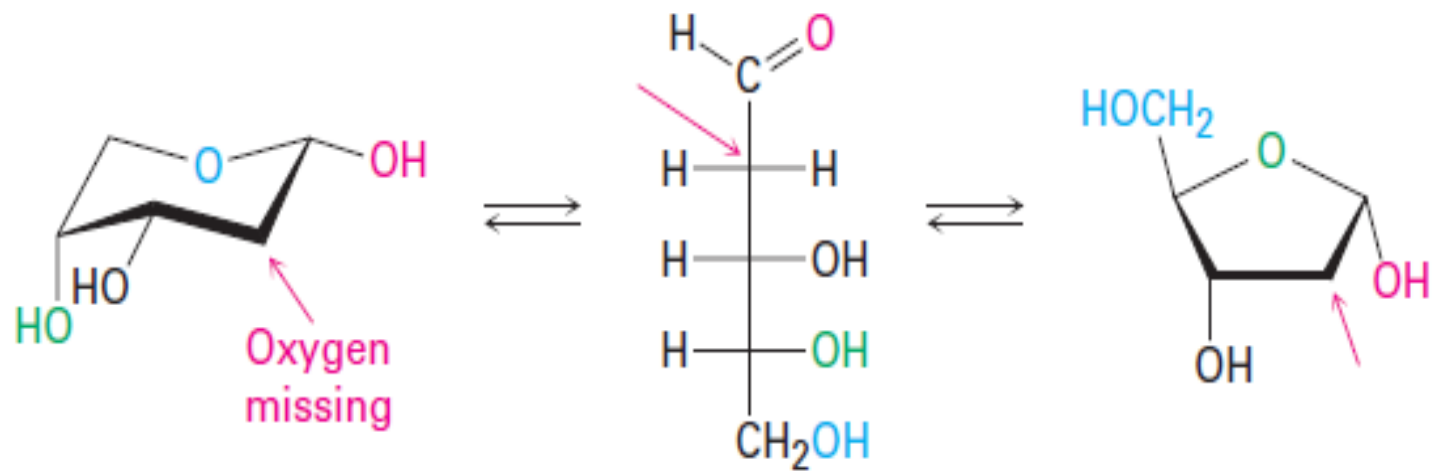


**α -L-Rhamnose
(6-deoxy-L-mannose)**



**α -L-Fucose
(6-deoxy-L-galactose)**

– 2-Deoxyribose



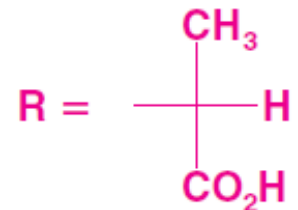
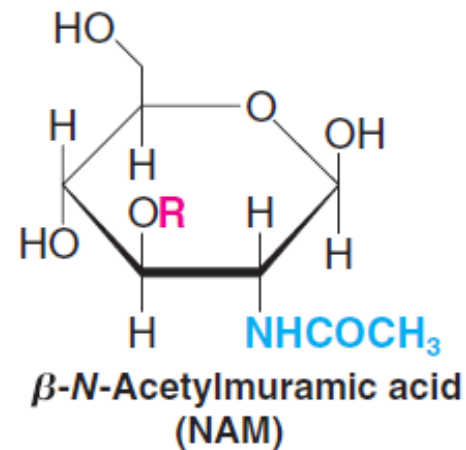
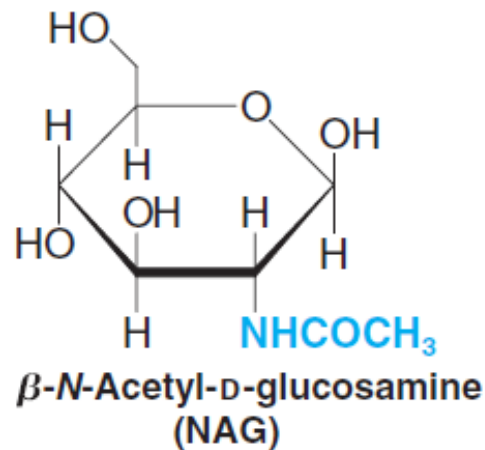
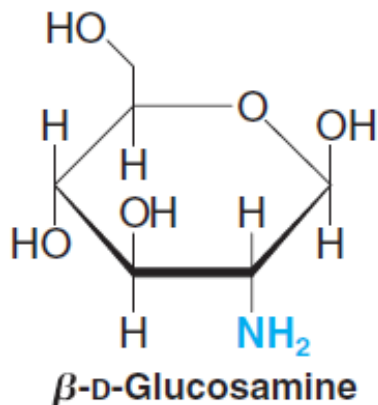
α -D-2-Deoxyribofuranose (40%)
(+ 35% β anomer)

(0.7%)

α -D-2-Deoxyribofuranose (13%)
(+ 12% β anomer)

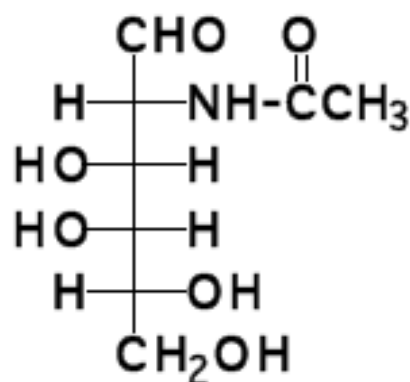
- **Amino sugars**

- Result from replacement of one or more hydroxyl groups in a sugar by **amino groups**.

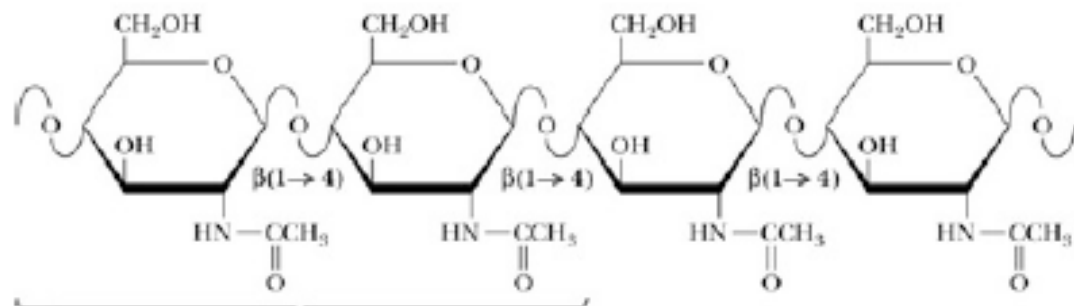


Structural Polysaccharides

- **Chitin:** the major structural component of the exoskeletons of invertebrates, such as insects and crustaceans; also occurs in cell walls of algae, fungi, and yeasts
 - composed of units of N-acetyl- β -D-glucosamine joined by β -1,4-glycosidic bonds

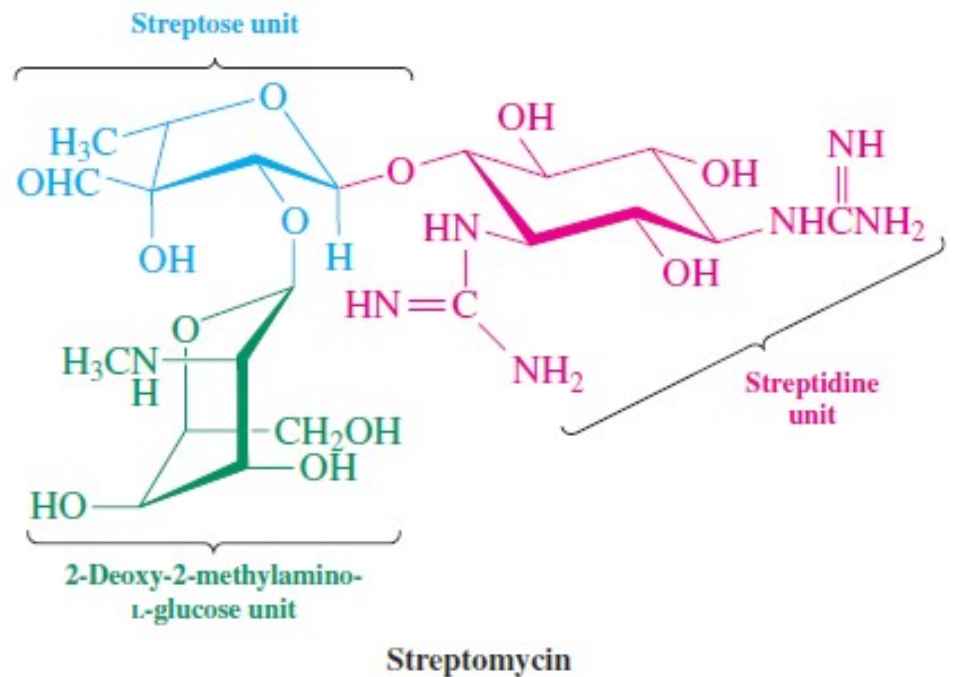
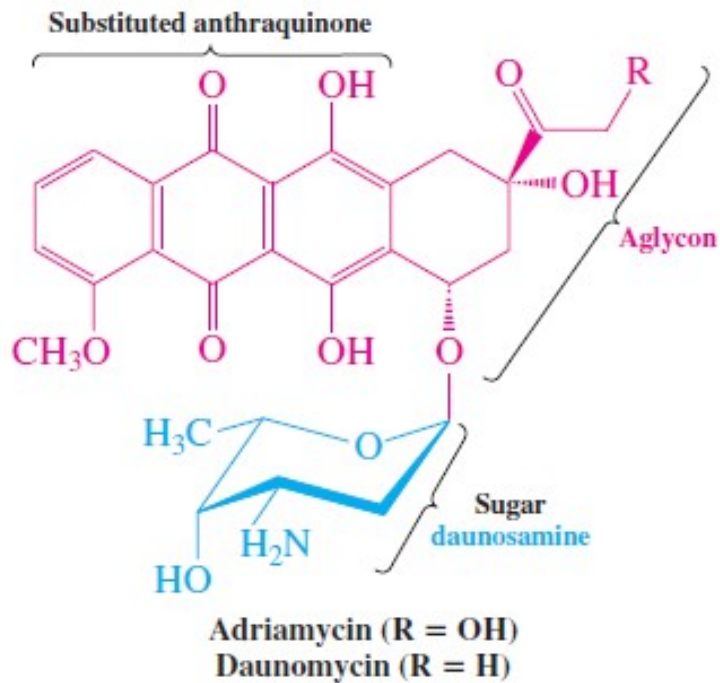


N-Acetyl-D-glucosamine



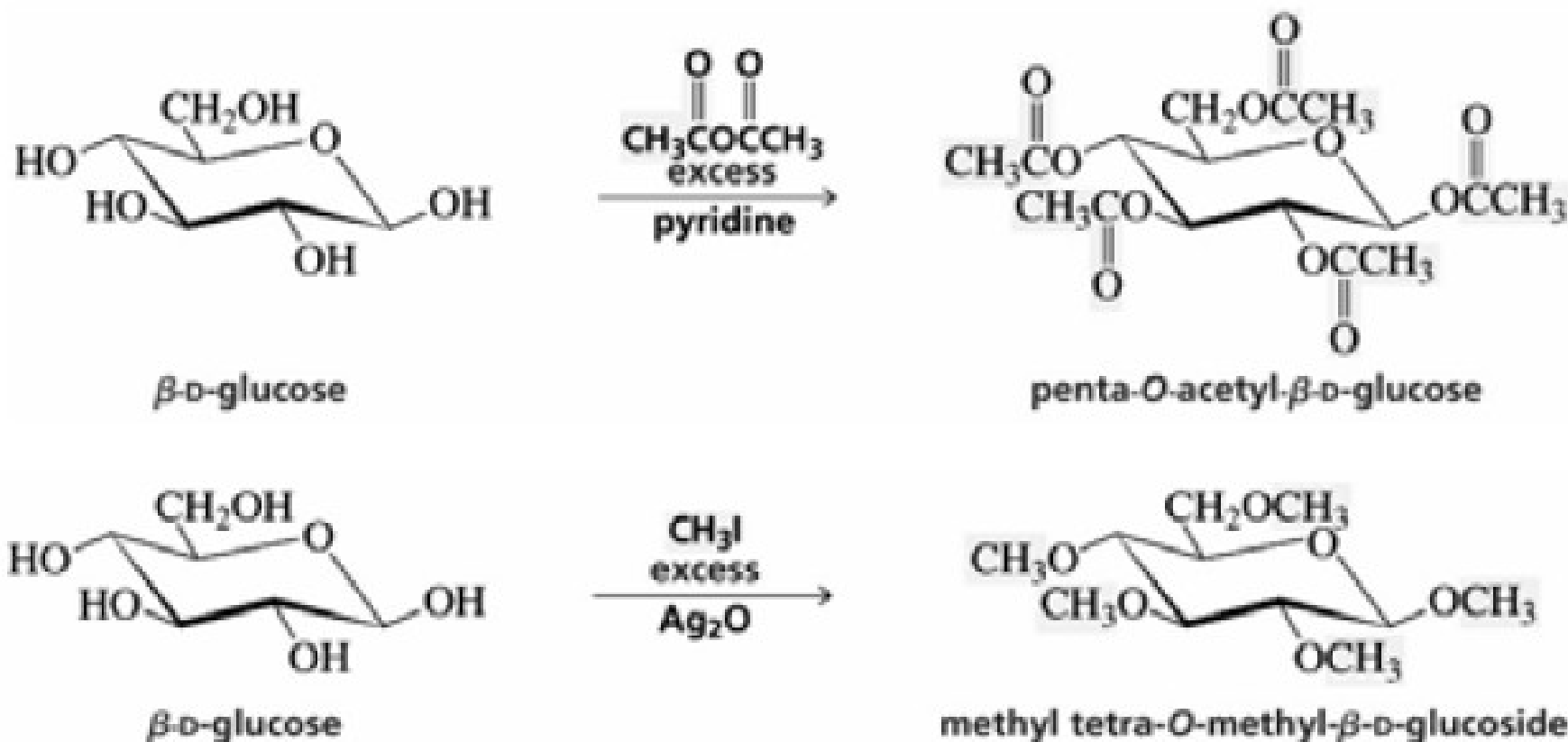
Repeating disaccharide
in chitin

– Adriamycin and daunomycin and aminoglycoside antibiotics



2.3.6. Reactions of monosaccharides

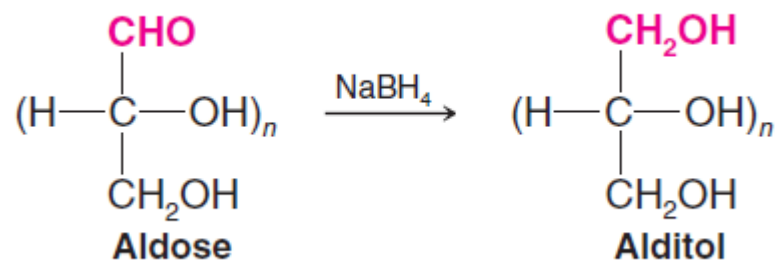
Acylation and Alkylation of Monosaccharides



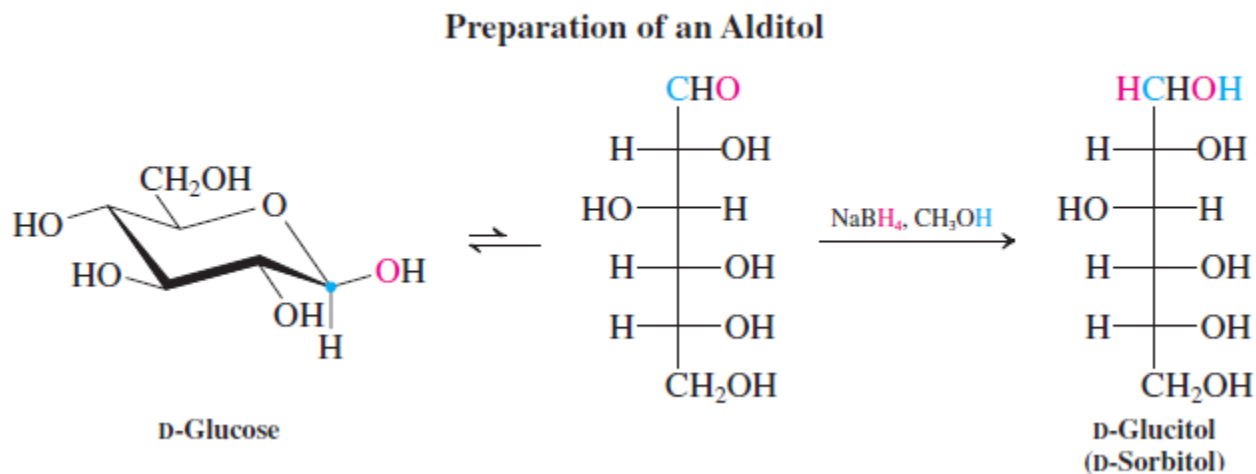
2.3.6.3. *Reduction*

- **Reduction**

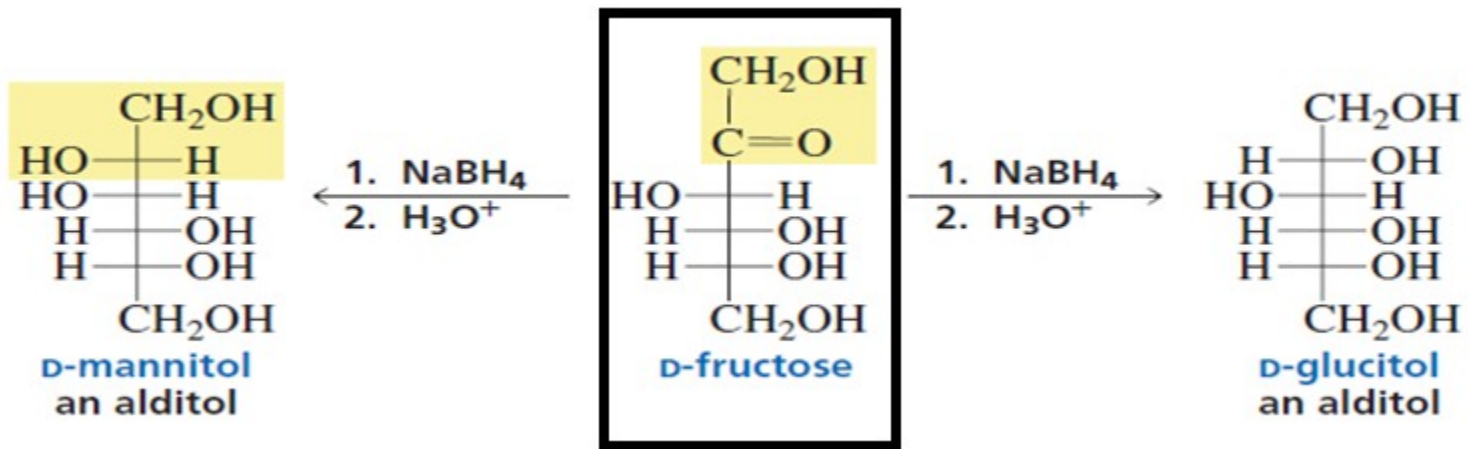
- Sodium borohydride (NaBH_4) to polyalcohols $\Rightarrow$ alditols



- Aldoses only one alditol

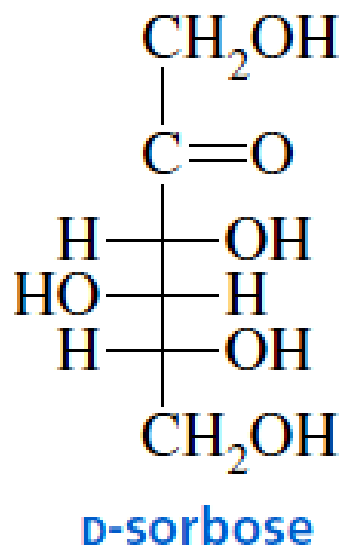
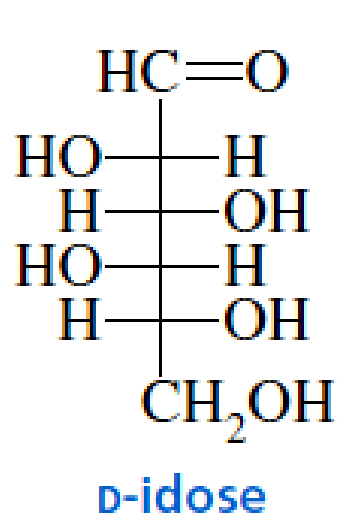


– Ketoses $\Rightarrow$ Two alditols. Why?



- **Exercise**

- What products are obtained from the reduction of



2.3.6.4. Oxidation

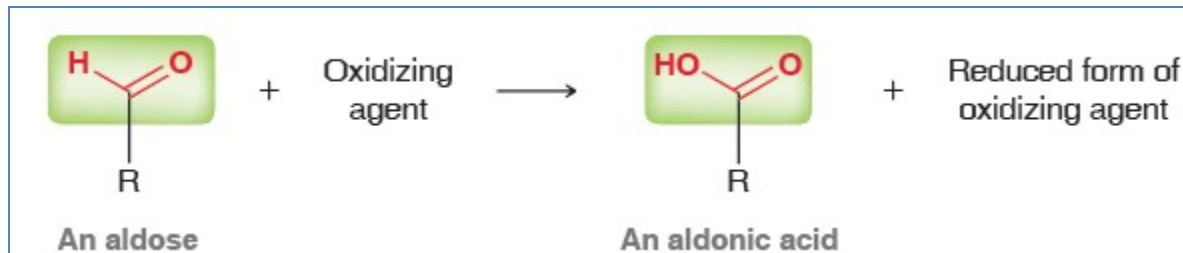
- **Monosaccharides**

- **Oxidations**

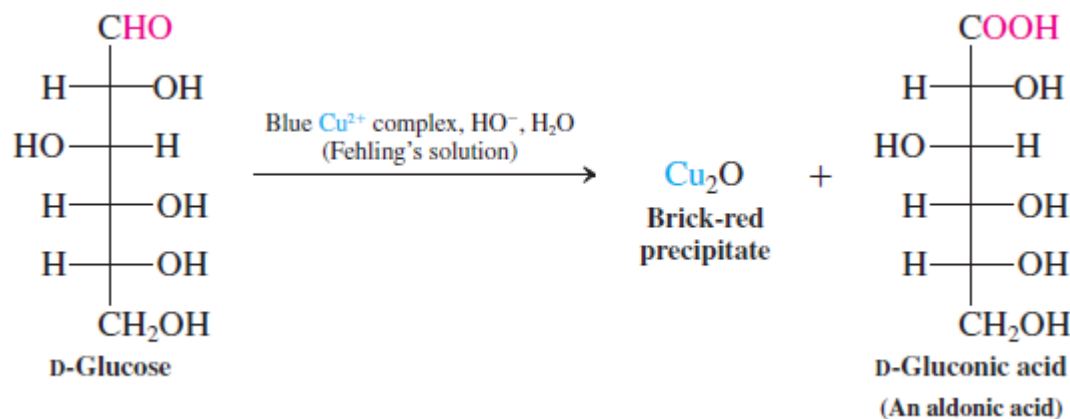
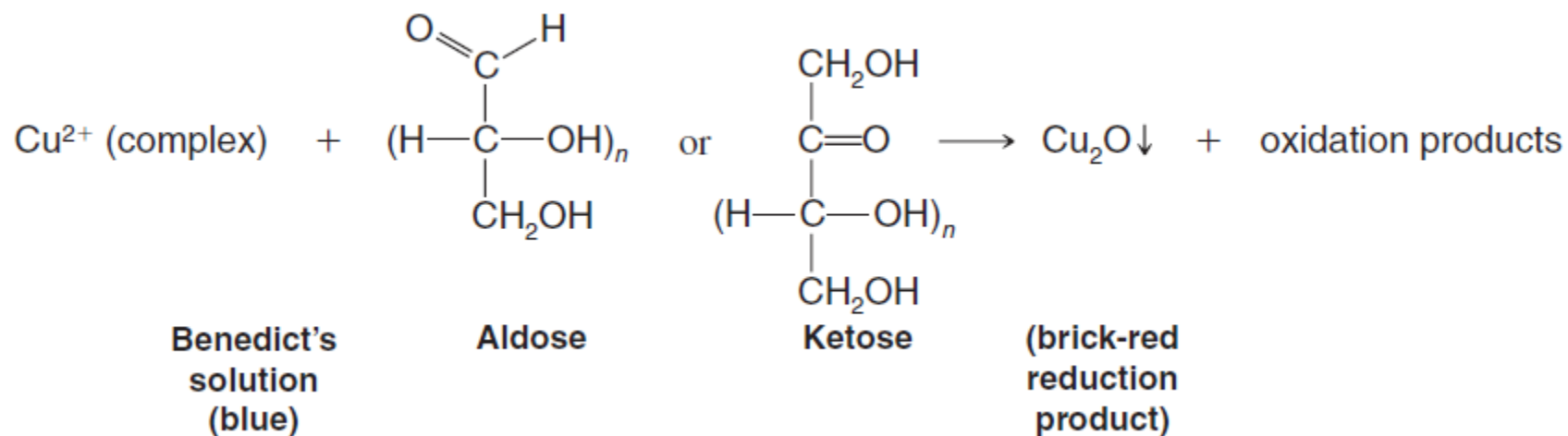
- a. Benedict's or Tollens' reagents
 - b. Bromine water
 - c. Nitric acid
 - d. Periodic acid

a. Benedict's or Tollens' Reagents: Reducing Sugars:

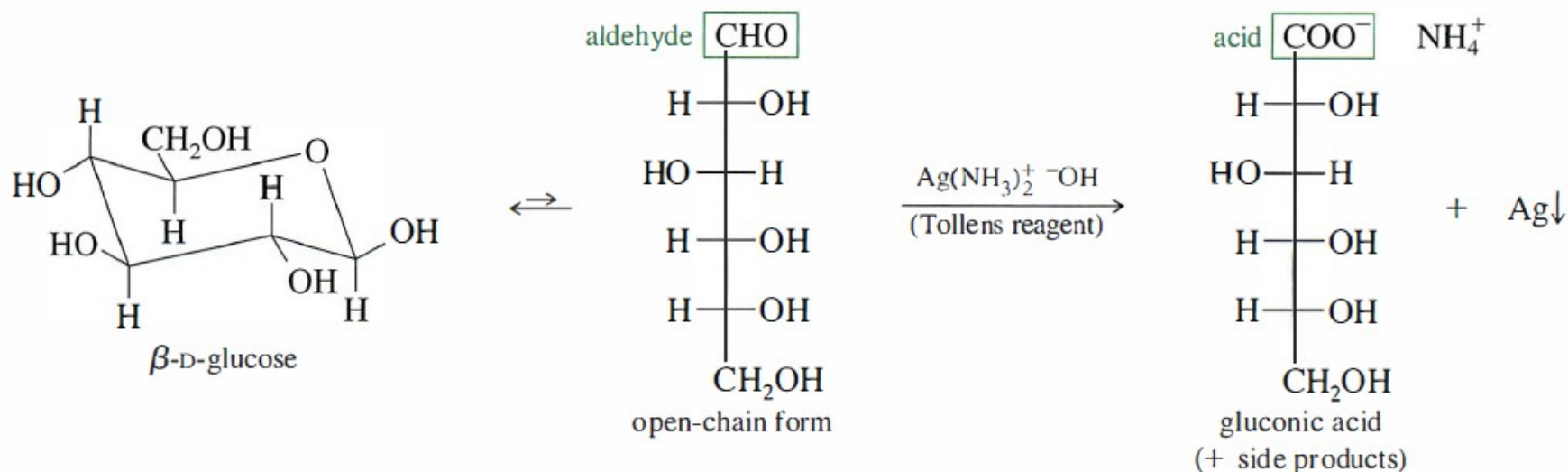
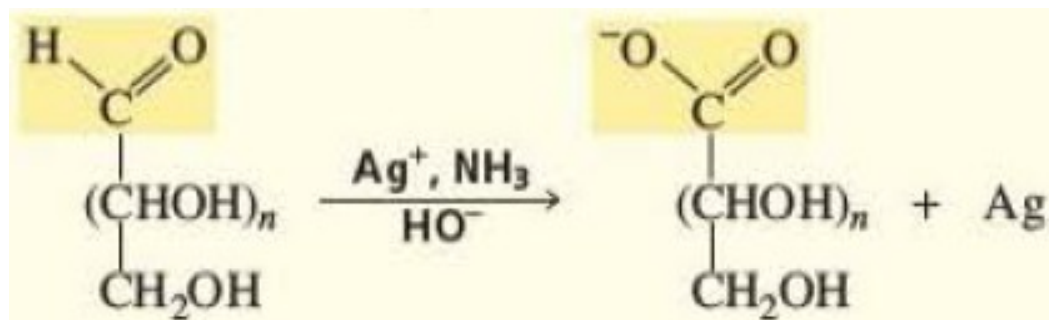
- Oxidize the carbonyl group to aldonic acids



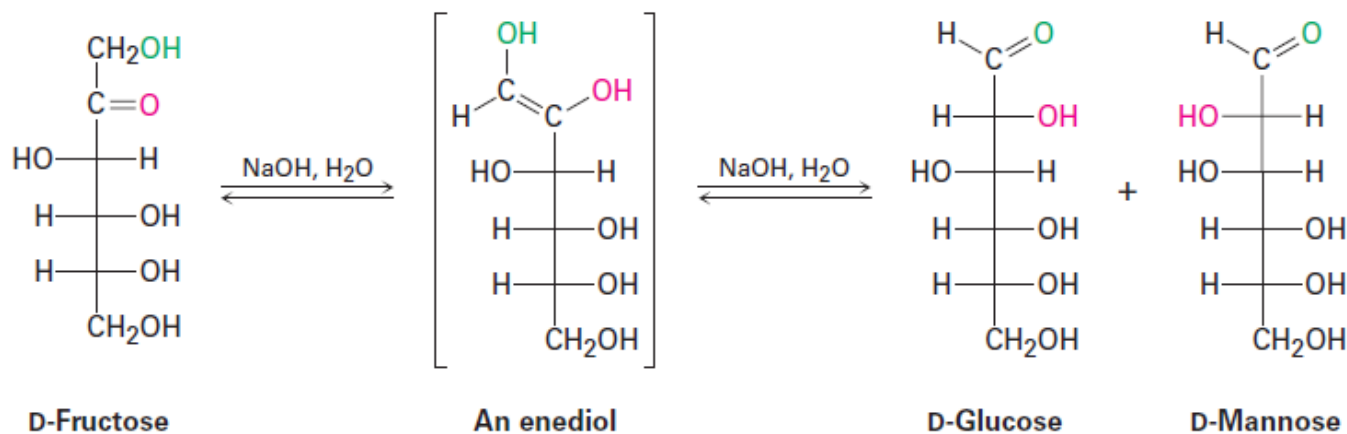
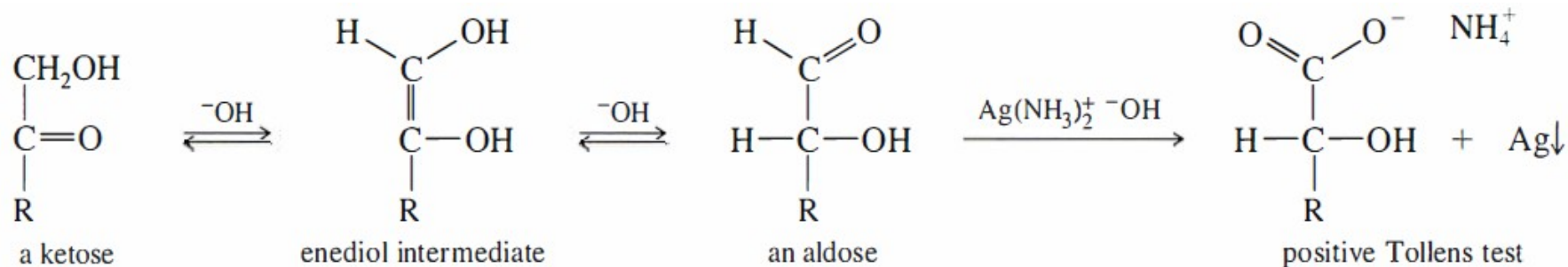
- Benedict's and Fehling's solution $\Rightarrow$ Cu^{2+}**



- Tollens reagent $\rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$

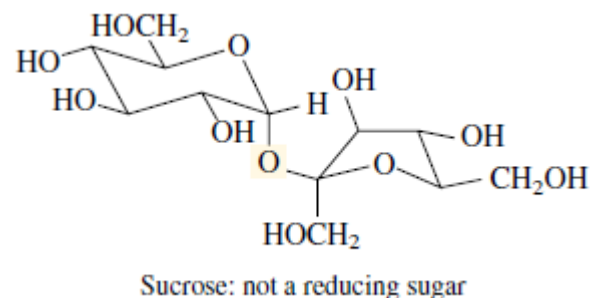
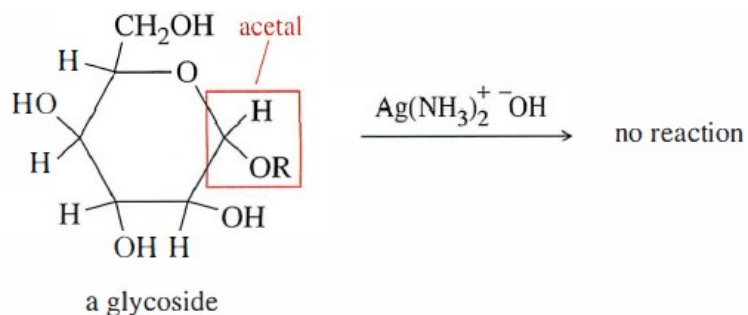
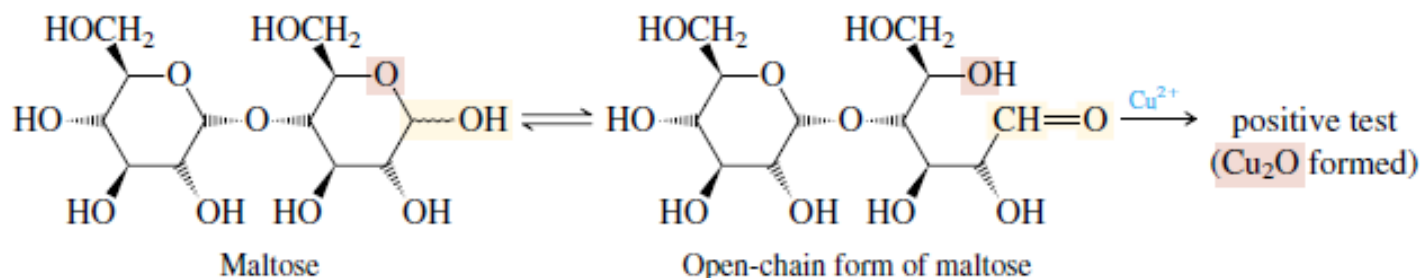


- Only oxidize only those containing aldehyde functional group.
 - Why, then, ketoses give positive test?



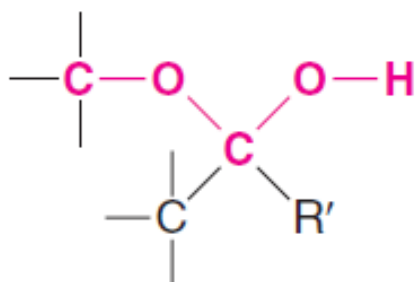
• Reducing sugars

- Positive test with Tollens' or Benedict's solutions
 - Only those containing **free** hemiacetal group or the open chain form of sugar
- Acetals or ketals $\Rightarrow$ Negative test



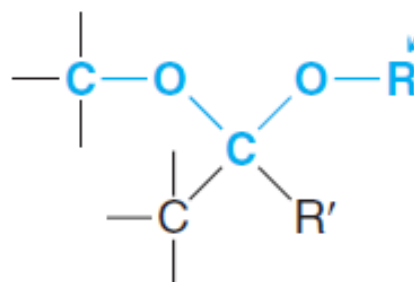
– Summary

Reducing Sugar



Hemiacetal ($R' = \text{H}$
or CH_2OH)
(gives positive
Tollens' or
Benedict's test)

Nonreducing Sugar



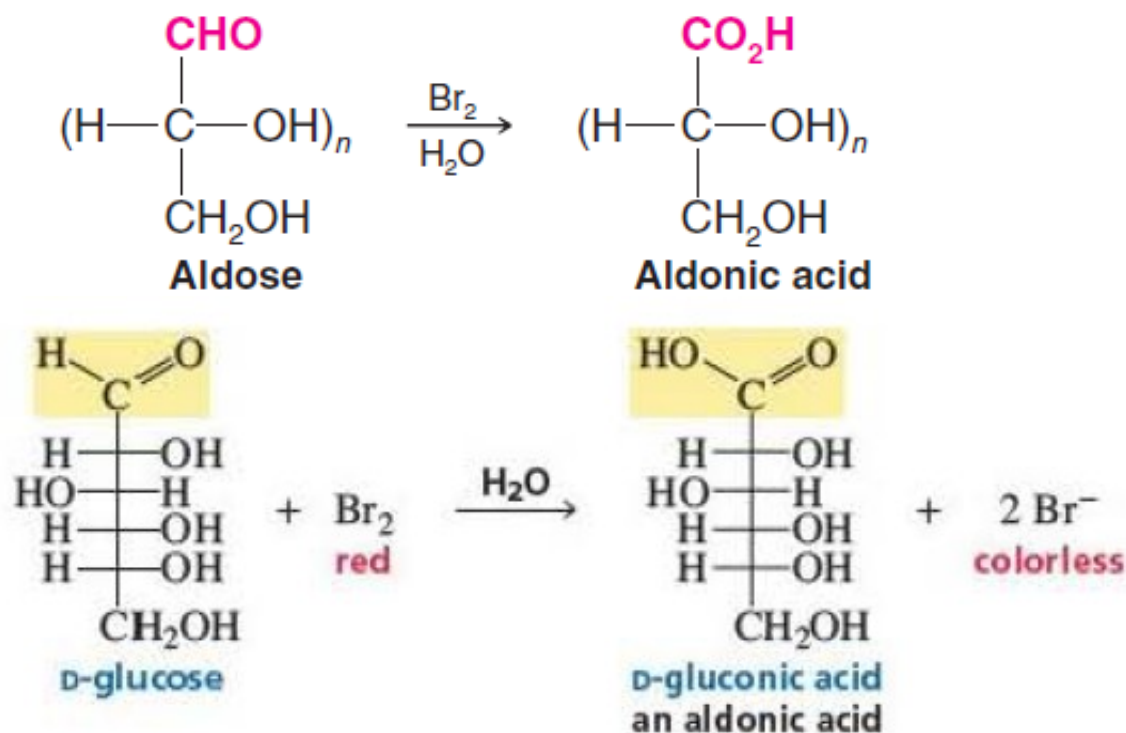
Alkyl group or
another sugar

Acetal ($R' = \text{H}$
or CH_2OH)
(does not give a
positive Tollens' or
Benedict's test)

b. Bromine water:

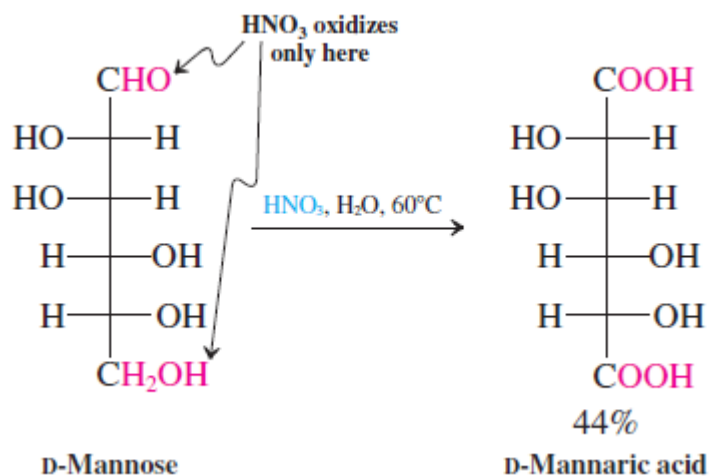
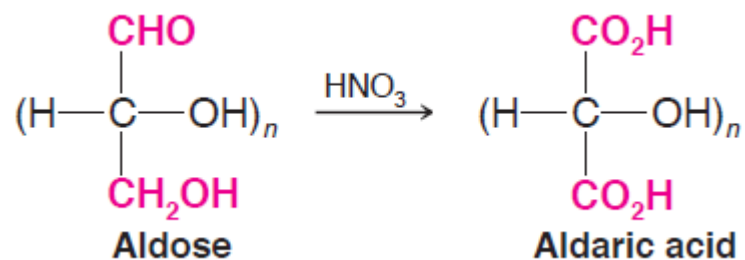
– Acidic reagent $\Rightarrow$ No isomerization

- Only aldoses
- Not strong enough to oxidize a ketose



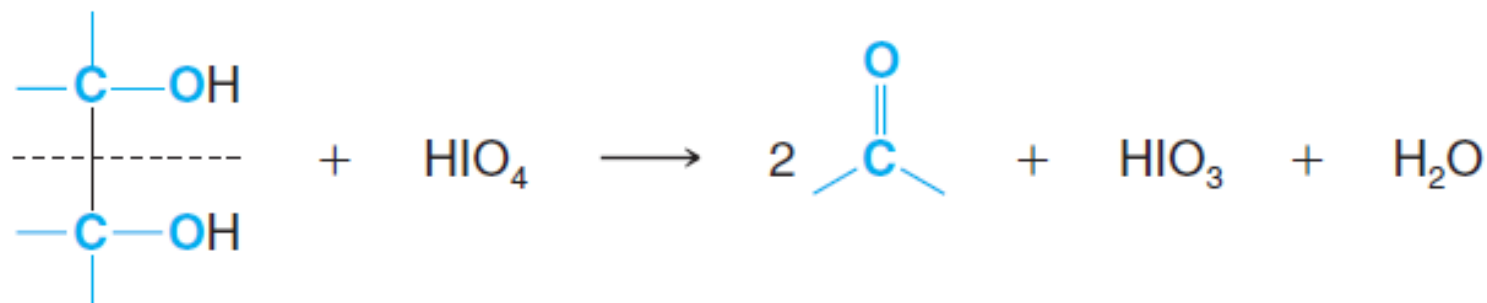
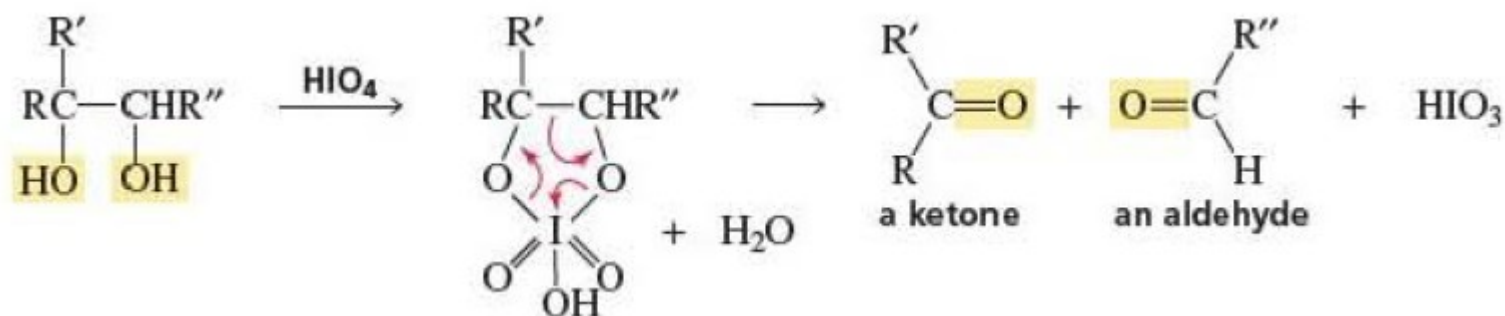
c. Nitric acid:

– Strong oxidizing agent



d. Periodate oxidations: Oxidative cleavage of polyhydroxy compounds:

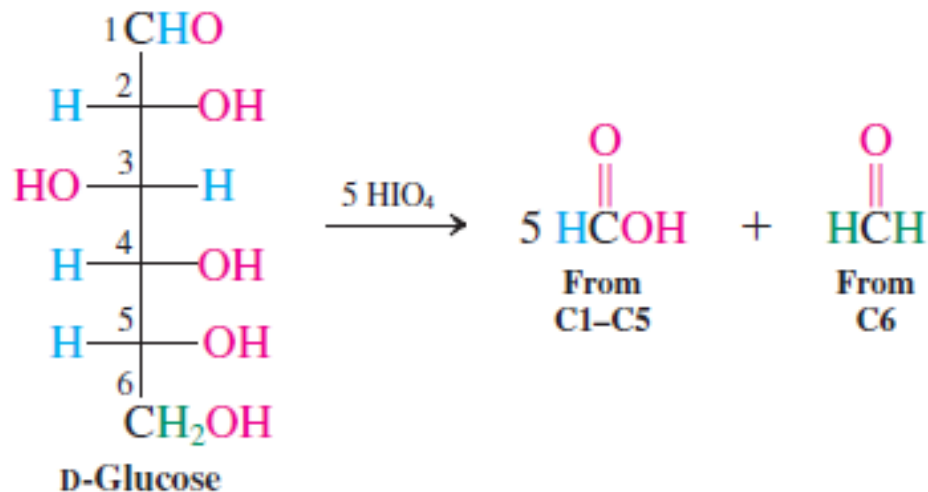
– Cleavage of vicinal diols



– Oxidative cleavage of sugars

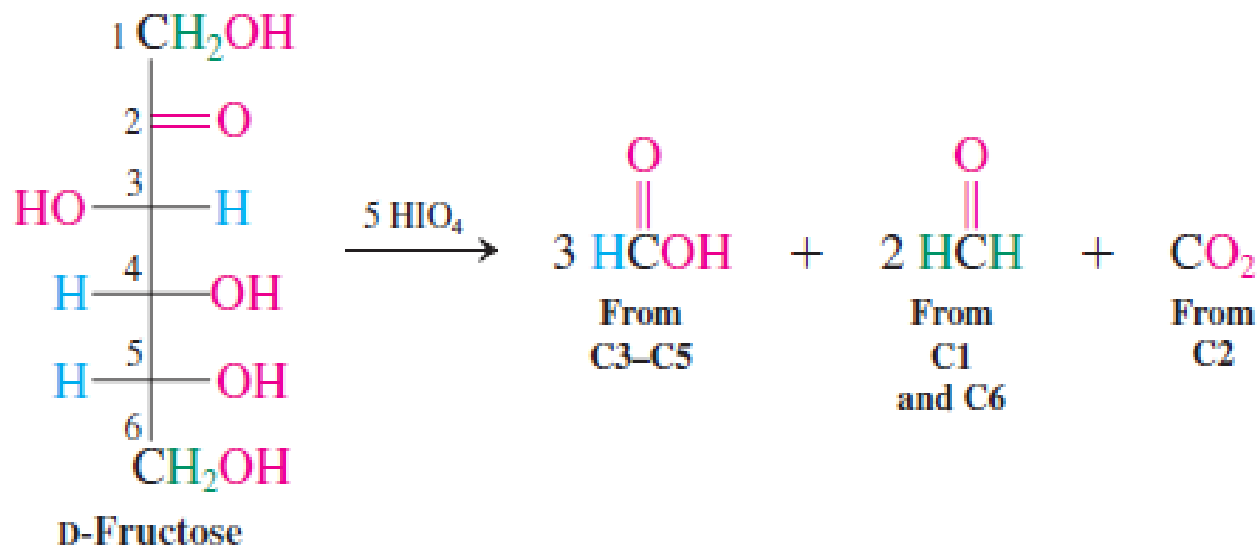
- formic acid, formaldehyde, and CO_2
- Facts
 - C-C cleavage $\Rightarrow$ 1 molecule of HIO_4
 - Aldehyde and 2° alcohol $\Rightarrow$ formic acid
 - 1° OH $\Rightarrow$ formaldehyde
 - Ketone $\Rightarrow$ CO_2

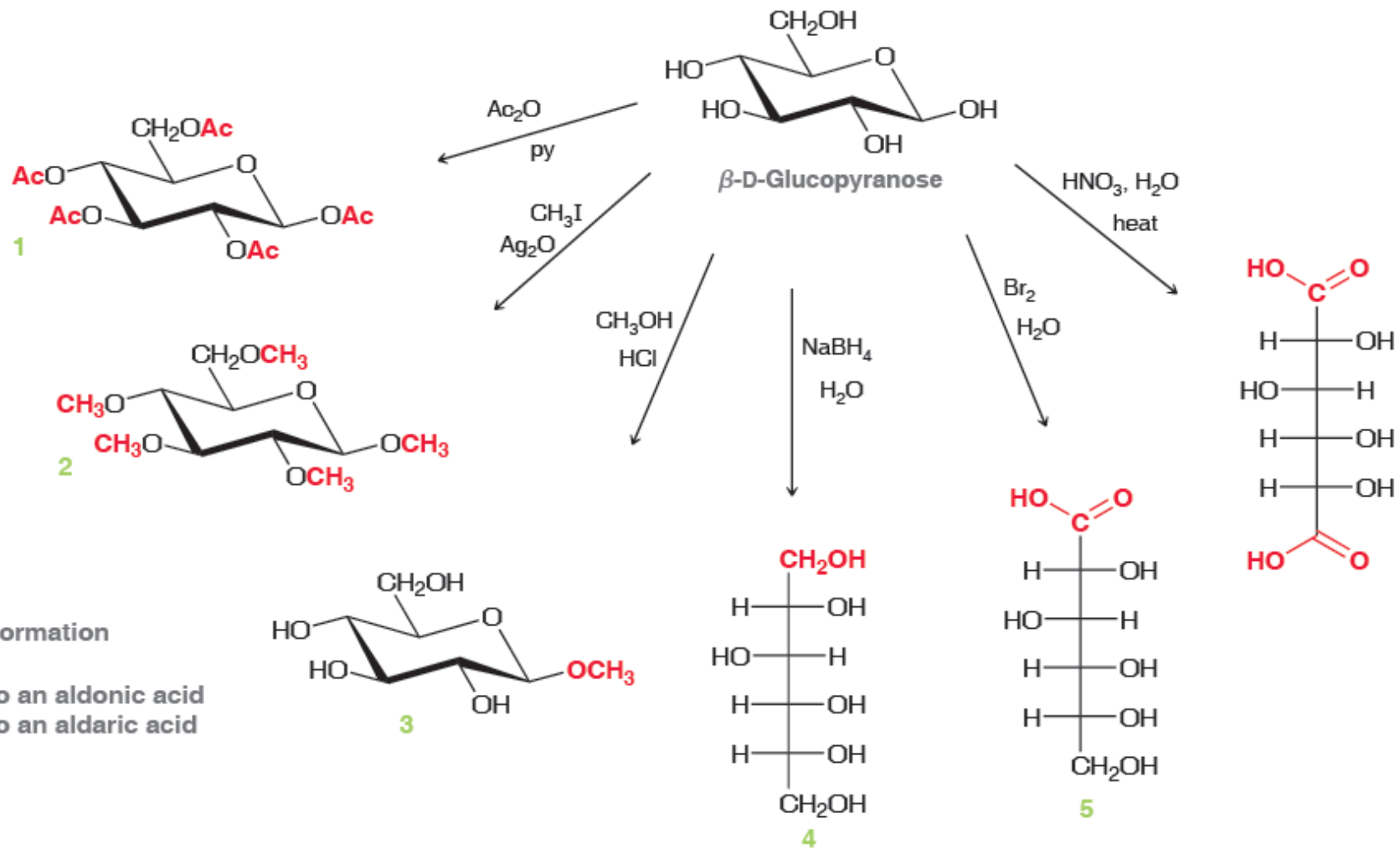
– Example:



- **Exercise**

- Predict the degradation products of fructose





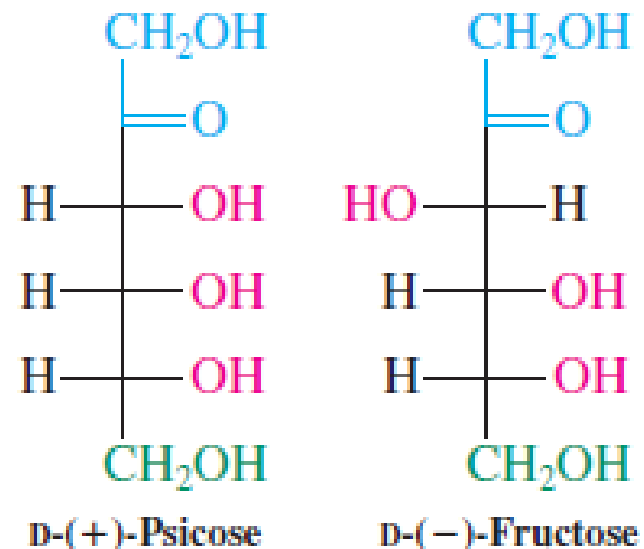
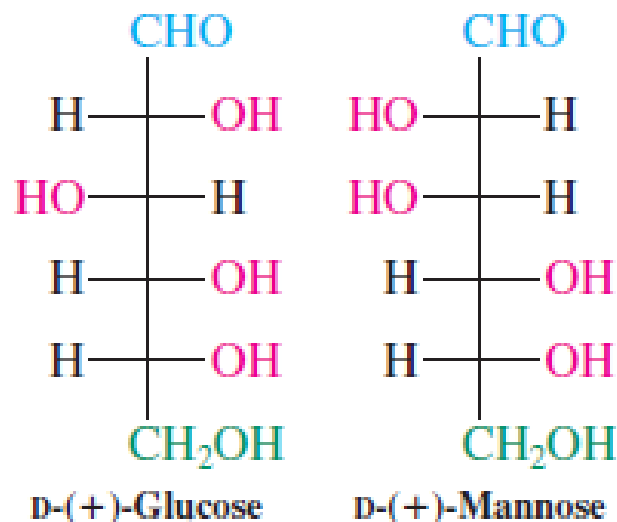
1. Acetylation
2. Alkylation
3. Glycoside formation
4. Reduction
5. Oxidation to an aldonic acid
6. Oxidation to an aldaric acid

2.3.6.1. The reactions of monosaccharides in
basic solutions:

Epimerization and isomerization

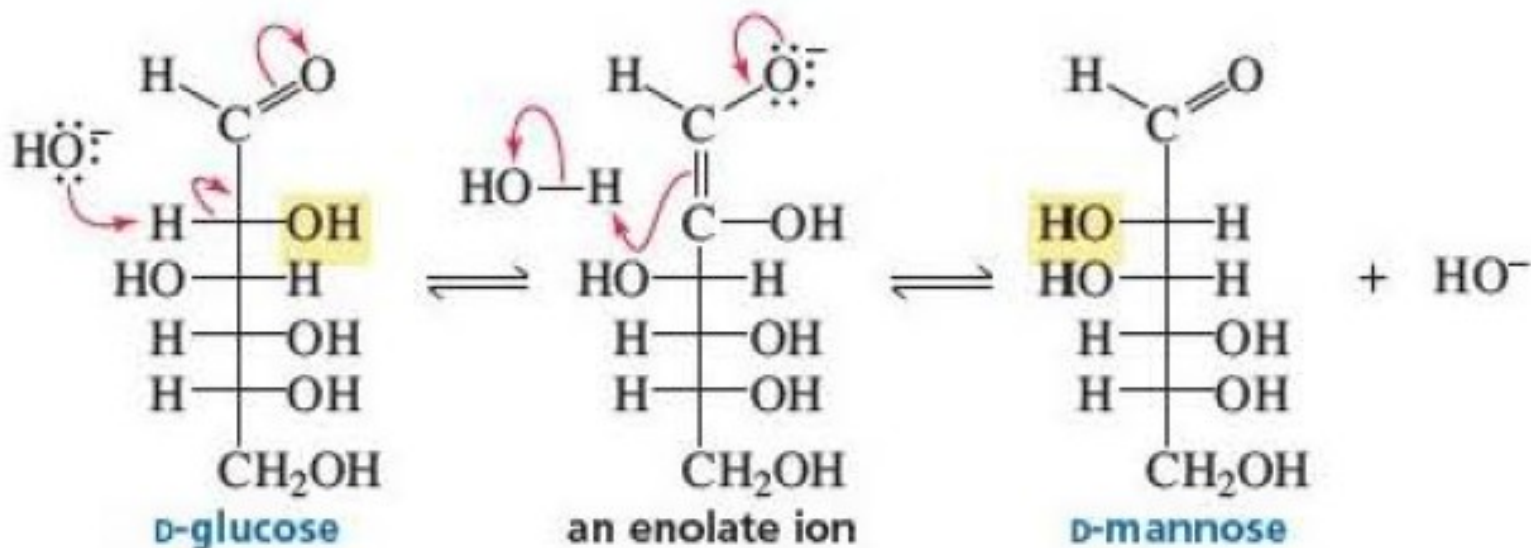
- **Epimers**

- Two diastereomeric sugars differing only in the configuration at a single asymmetric carbon atom.
- The interconversion of epimers is called **epimerization**.



- In basic solution monosaccharides undergo
 - Epimerization

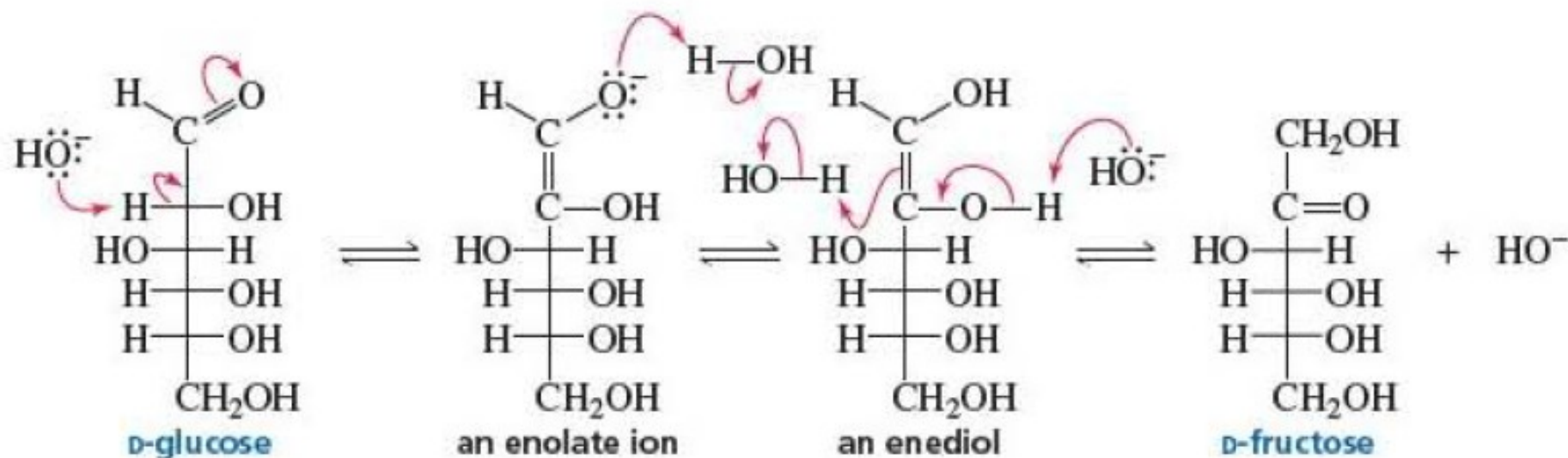
MECHANISM FOR THE BASE-CATALYZED EPIMERIZATION OF A MONOSACCHARIDE



ii. Isomerization

- Ketones
- Enediol rearrangement

MECHANISM FOR THE BASE-CATALYZED ENEDIOL REARRANGEMENT OF A MONOSACCHARIDE



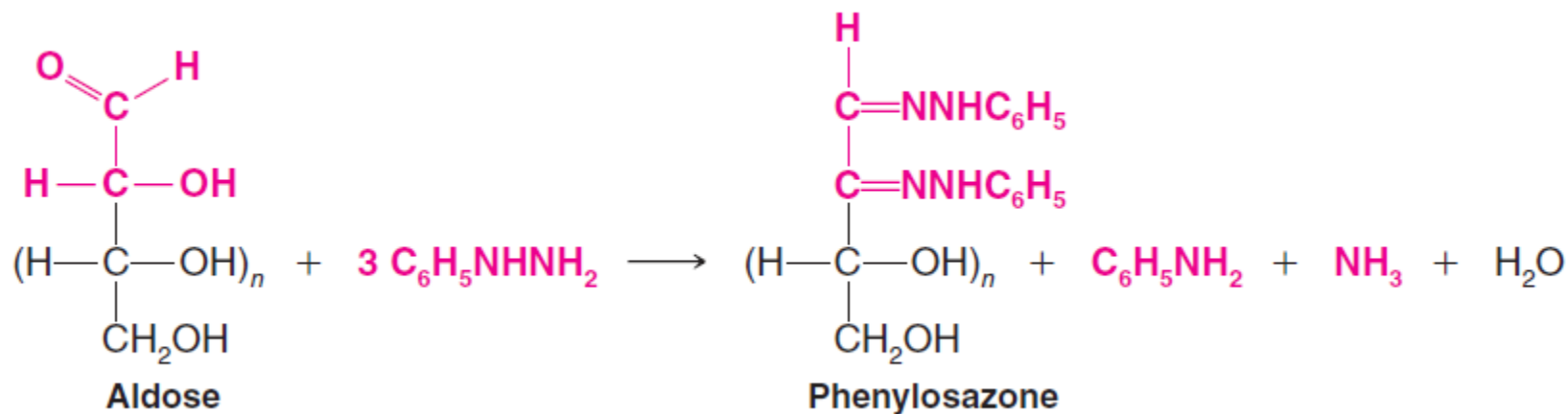
2.3.6.4. Osazone test:
Reaction (reduction) with
phenylhydrazine: Epimers

- **Osazone test:**

- Reaction of monosacharides with phenylhydrazine

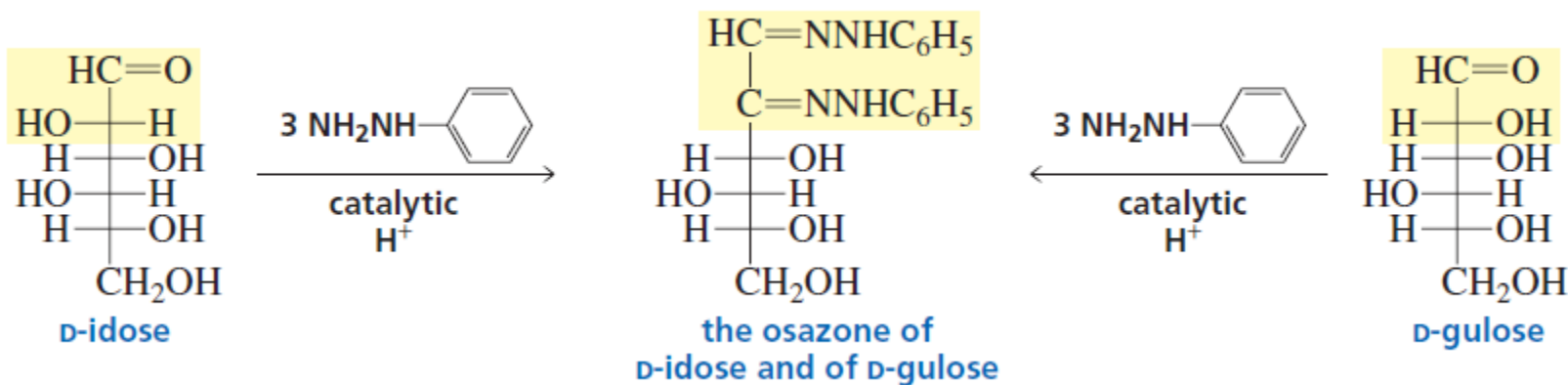
⇒ Osazones

- 1,2-bisarylhydrazone formed by reaction of an aldose or ketose with three molar equivalents of an arylhydrazone

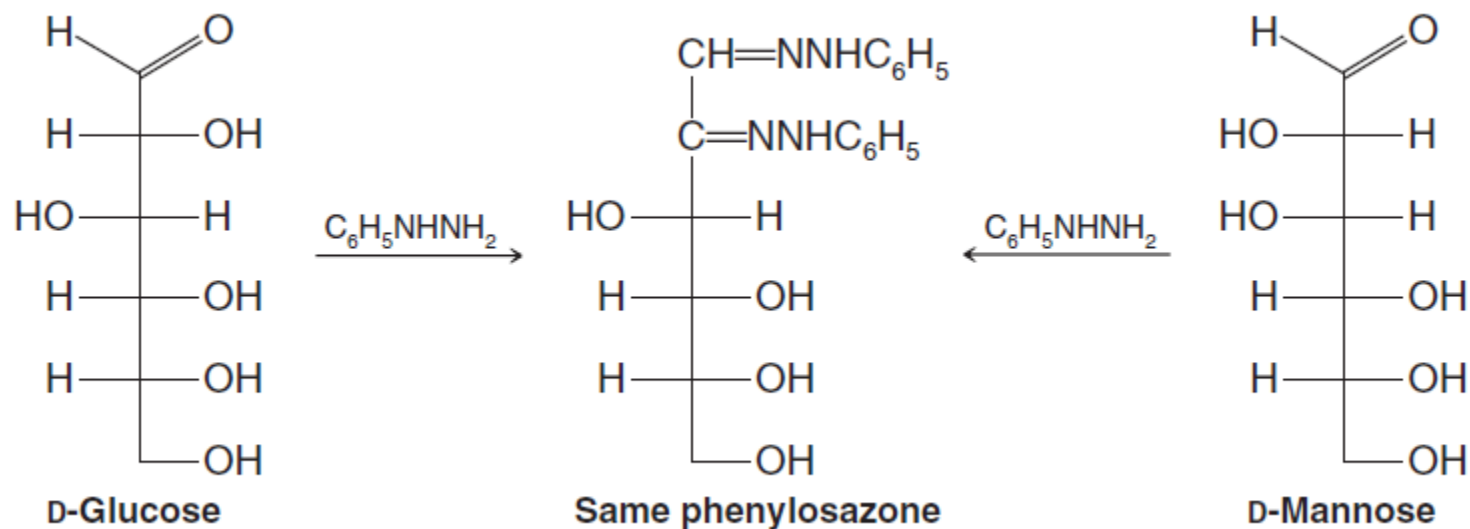


– Leads to loss of configuration at C-2

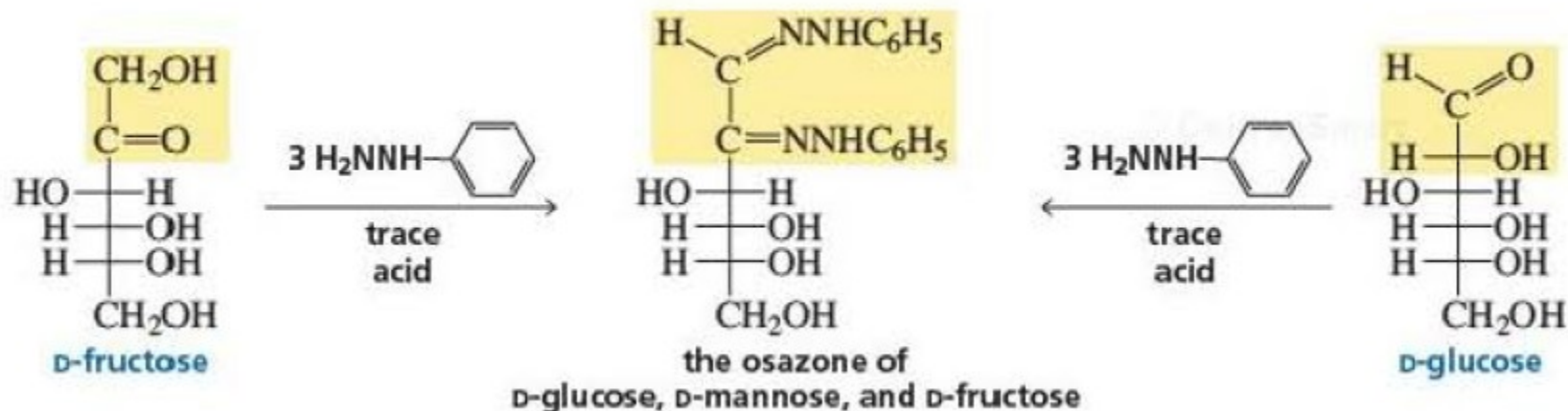
- Same osazones



- **Emil Fisher**
 - D-glucose and D-mannose

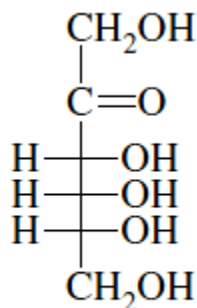


- C-1 and C-2 of ketoses $\Rightarrow$ Phenyllosazones

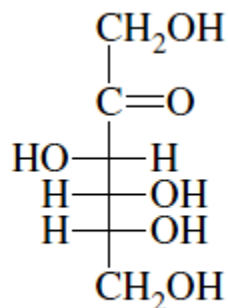


- **Exercise**

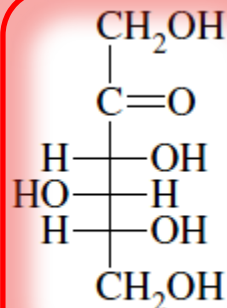
- What monosaccharides form the same osazone as D-sorbose?



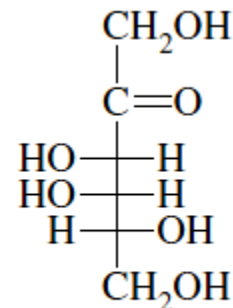
D-psicose



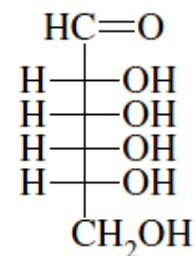
D-fructose



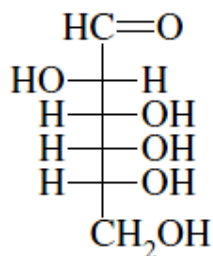
D-sorbose



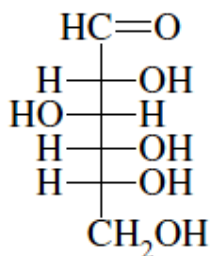
D-tagatose



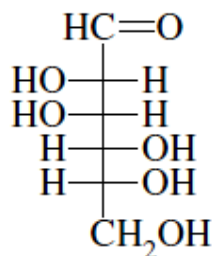
D-allose



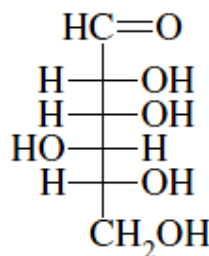
D-altrose



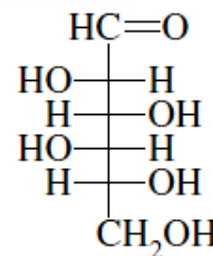
D-glucose



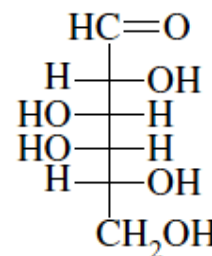
D-mannose



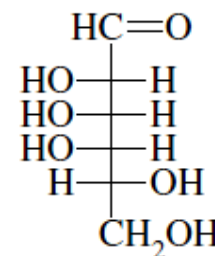
D-gulose



D-idose



D-galactose



D-talose

2.3.6.5. Kiliani–Fischer synthesis: Chain elongation

- **Kiliani–Fischer Synthesis**

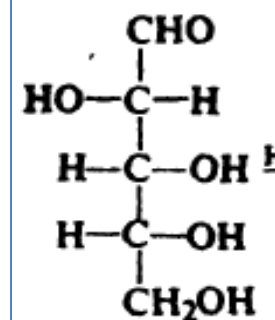
- Formation of higher monosaccharides ($n+1$) from lower ones (n)
- Two sugars
 - With different stereochemistry at the original carbonyl carbon



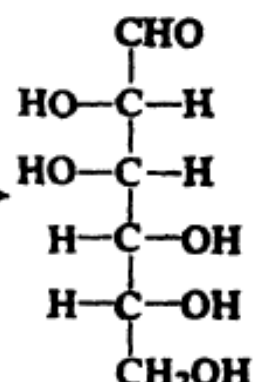
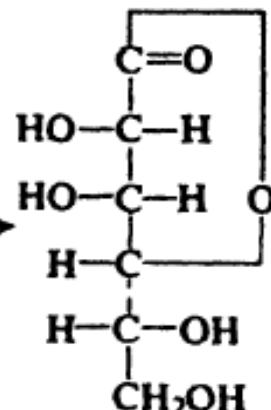
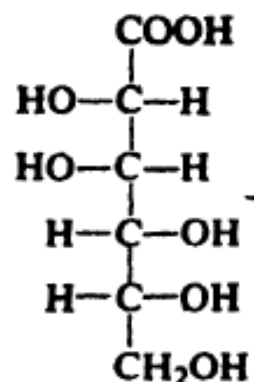
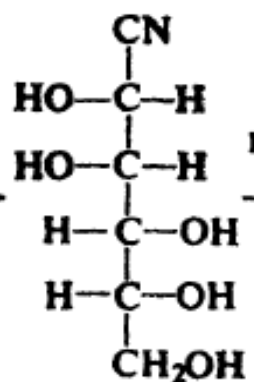
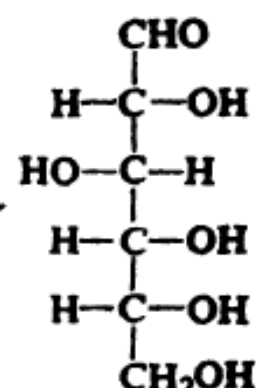
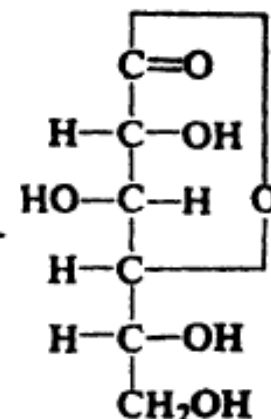
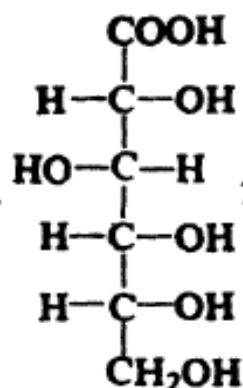
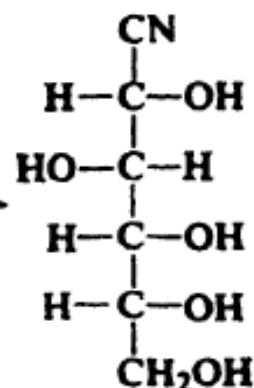
**Heinrich
Kiliani
(1855 - 1945)**



**Emil Fischer
(1852 – 1919)**



An aldopentose



Diastereomeric
cyanohydrins

Diastereomeric
aldonic acids

*Mixture usually
separated at
this point*

Diastereomeric
aldonolactones

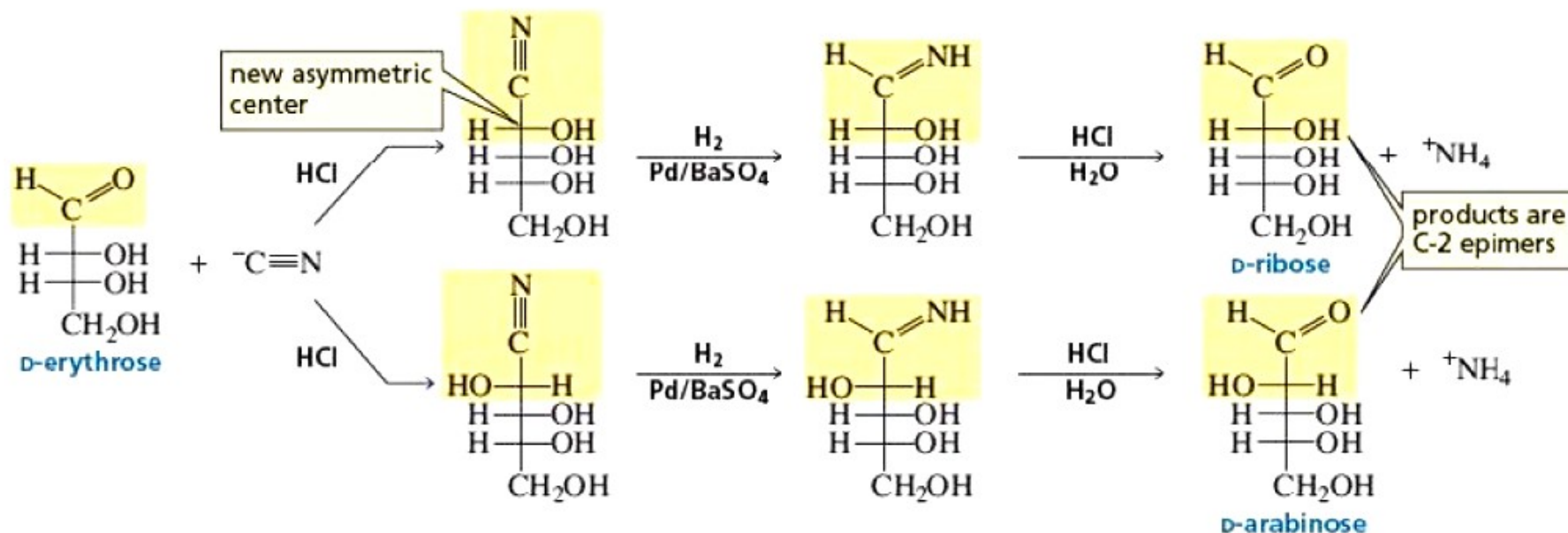
Diastereomeric
aldohexoses

Epimers

• Modified Kiliani–Fischer synthesis

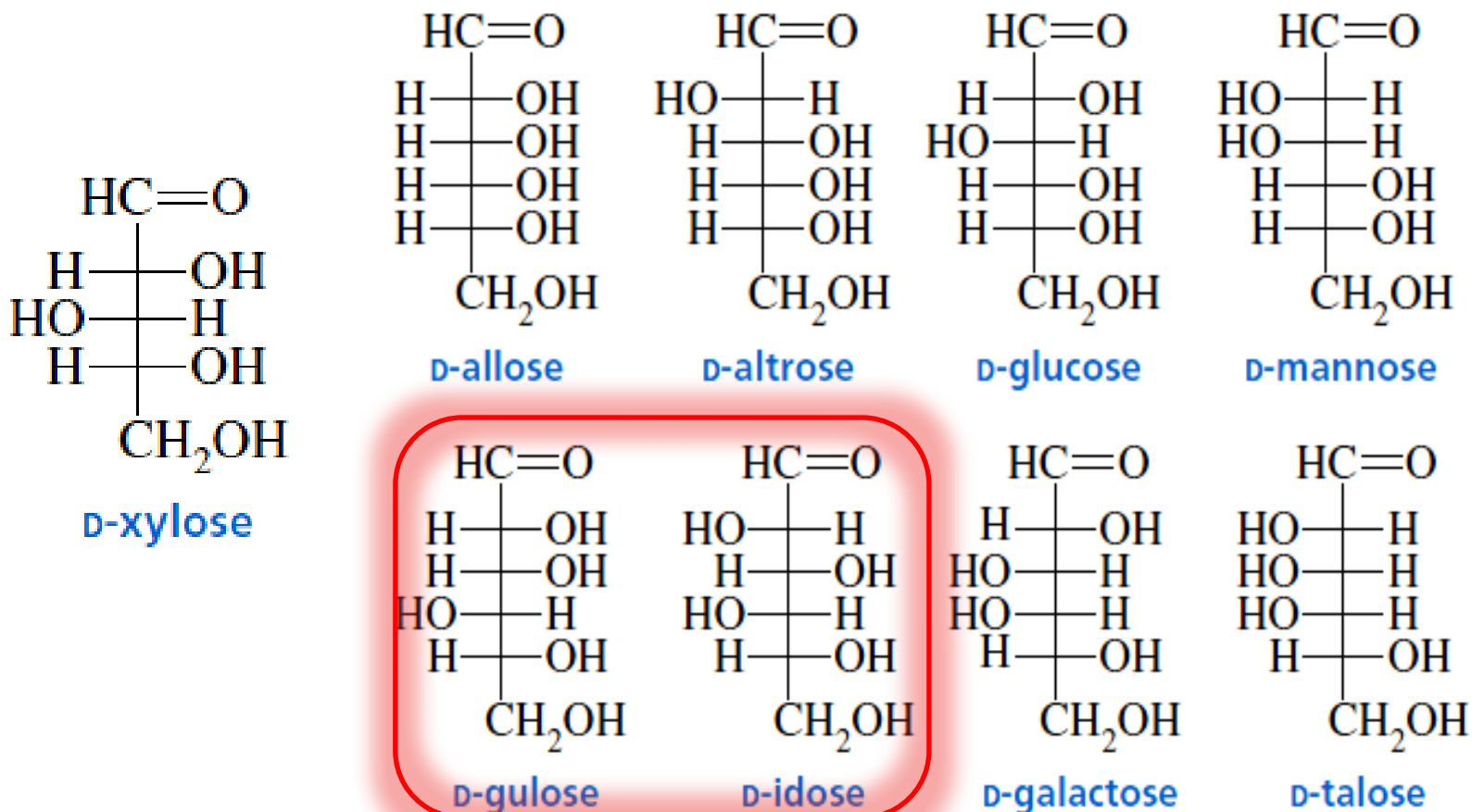
- Serianni and Barker in 1979
- Cyanohydrins to aldoses
 - Palladium (on barium sulfate) instead of Na (Hg)

the modified Kiliani–Fischer synthesis



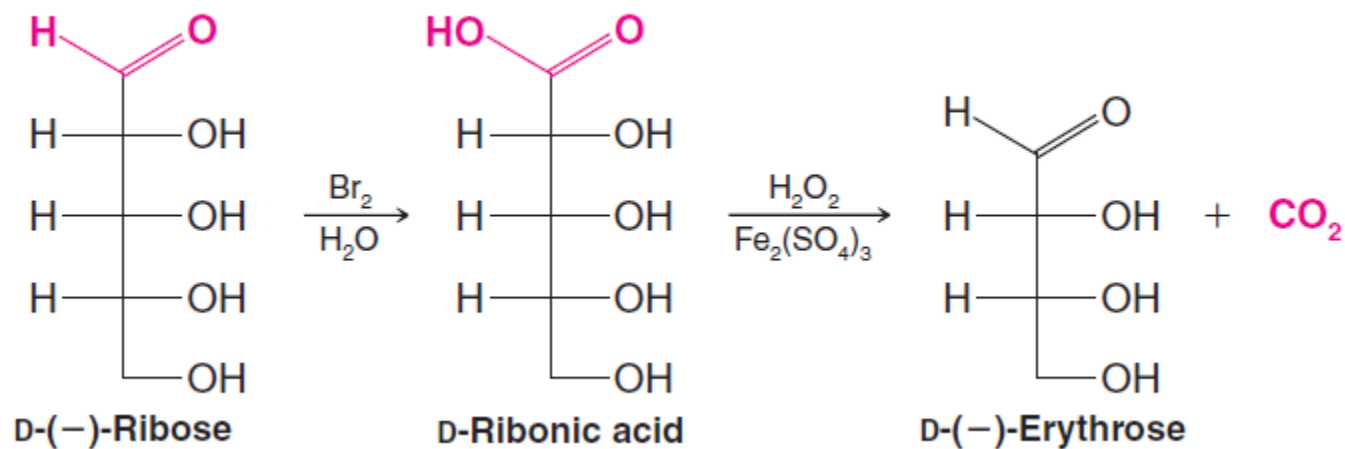
- **Exercise:**

- What monosaccharides would be formed in a Kiliani–Fischer synthesis starting with D-xylose?



2.3.6.5. *Chain shortening*

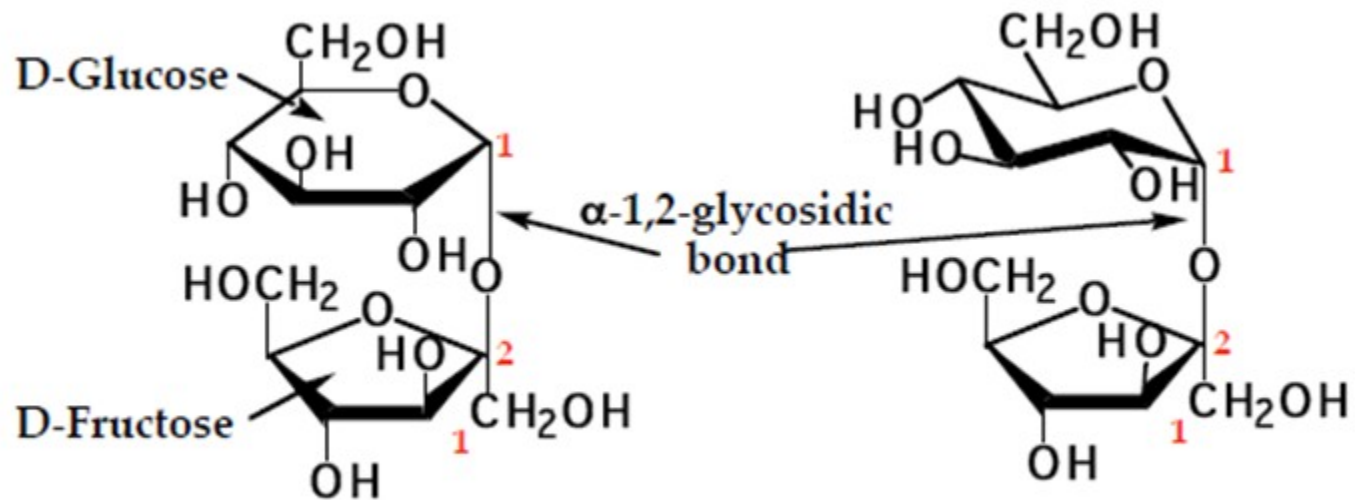
The Ruff Degradation



2.4. Disaccharides

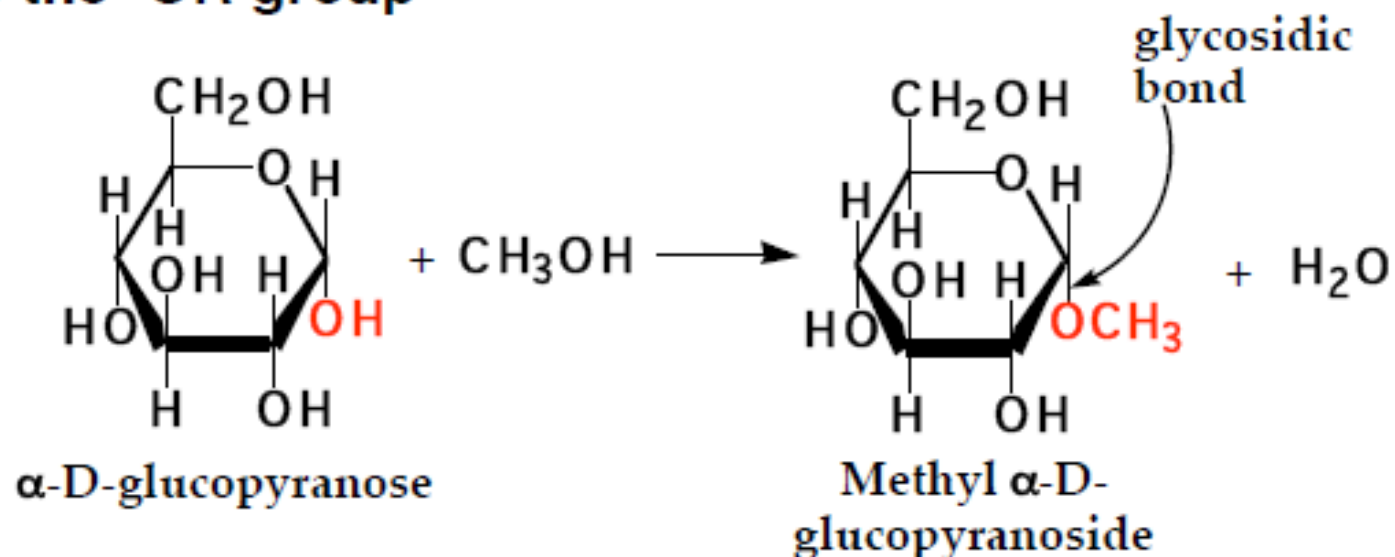
- **Disaccharides**

- glycosides formed from two monosacharides
- Connected by glycosidic linkages
- So, what are glycosides?



Formation of Glycosides

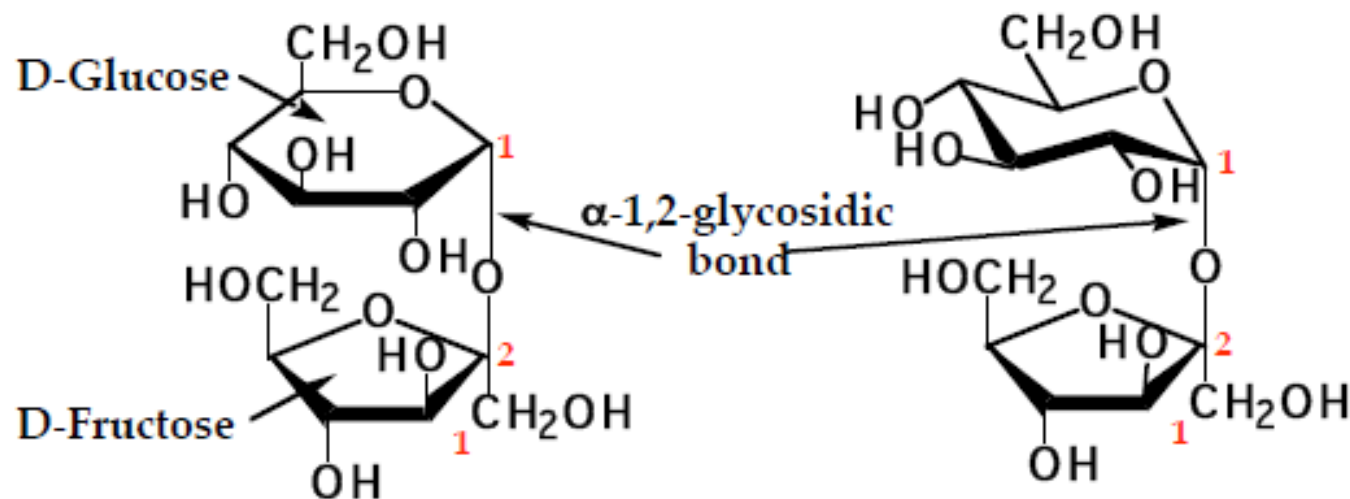
- **Glycoside**: a carbohydrate in which the -OH of the anomeric carbon is replaced by -OR
 - those derived from furanoses are **furanosides**; those derived from pyranoses are **pyranosides**
 - **glycosidic bond**: the bond from the anomeric carbon to the -OR group



Disaccharides

- **Sucrose**

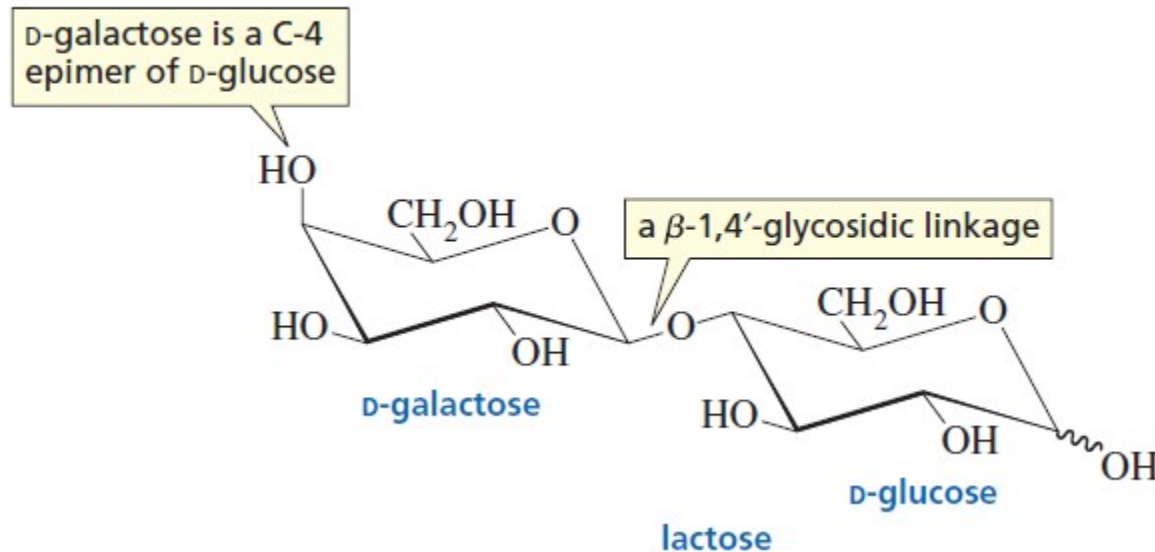
- table sugar; obtained from the juice of sugar cane and sugar beet
- one unit of D-glucose and one unit of D-fructose joined by an α -1,2-glycosidic bond



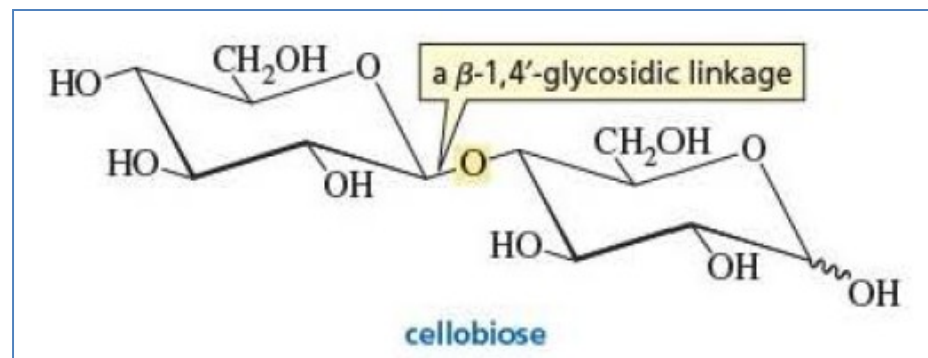
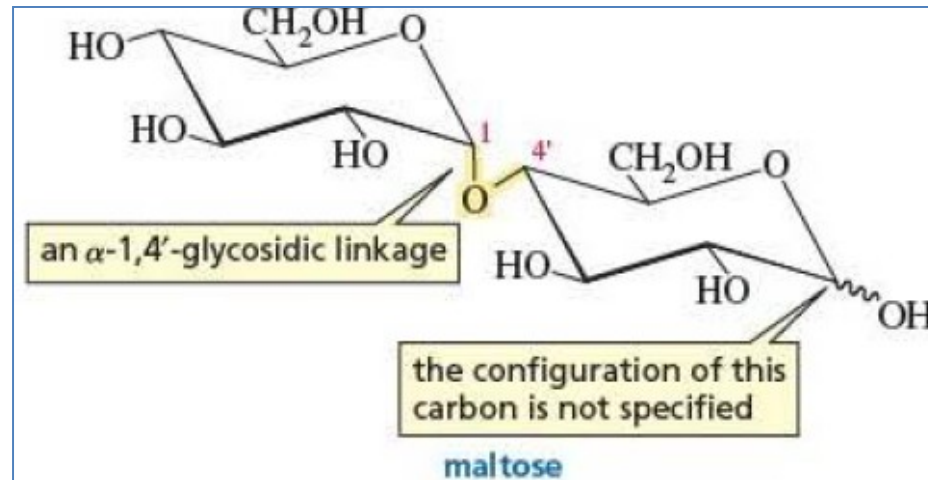
2.4.3. Lactose, Maltose and Cellobiose

- **Lactose**

- 4-*O*-(β -D-galactopyranosyl)- β -D-glucopyranose
- Product of β -1,4-glycosidic bond link of D-galactose and D-glucose



- **Maltose and cellobiose**
 - Isomeric dimers of D-glucose



2.6. Polysaccharides

- **Polysaccharides**

- Polymers of > 10 monosaccharides

- Property

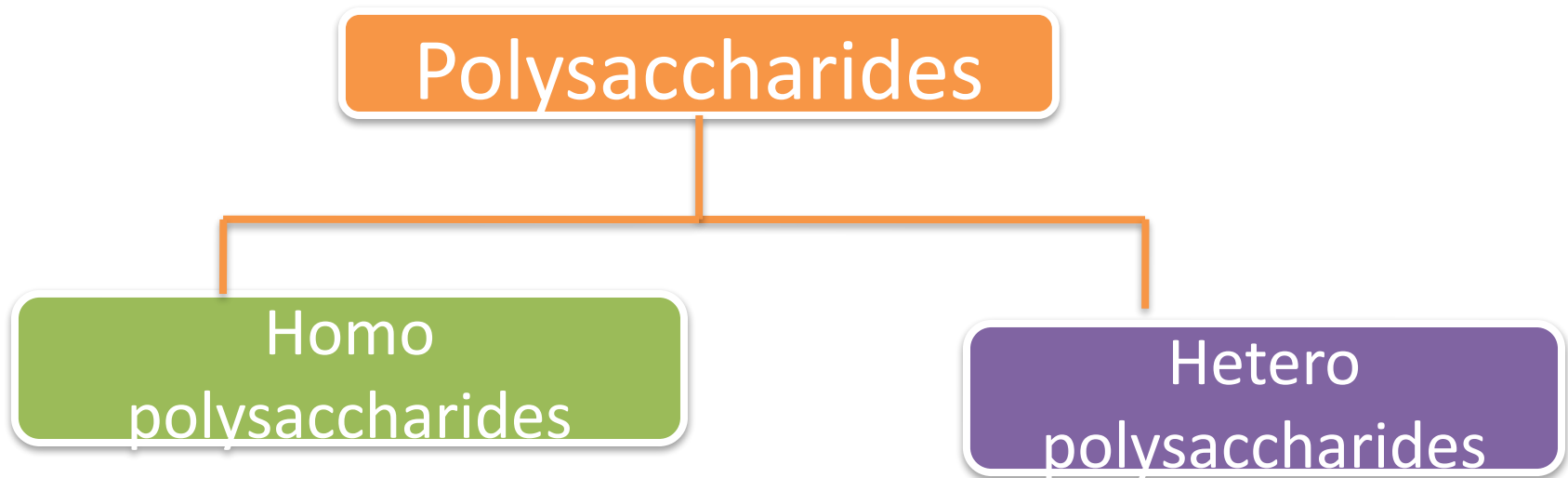
- *species of monosaccharides*
 - *types of linkages* between residues
 - *extent and type of chain branching*

2.6.1. Classification of polysaccharides

A. Based on source

- Plant polysaccharides
- Animal polysaccharides
- Algal polysaccharides,
- Bacterial polysaccharides, etc

B. Based on composition



- Homo polysaccharides- gives single monosaccharide units on hydrolysis
- Hetero polysaccharides- gives more than one monosaccharides units on hydrolysis

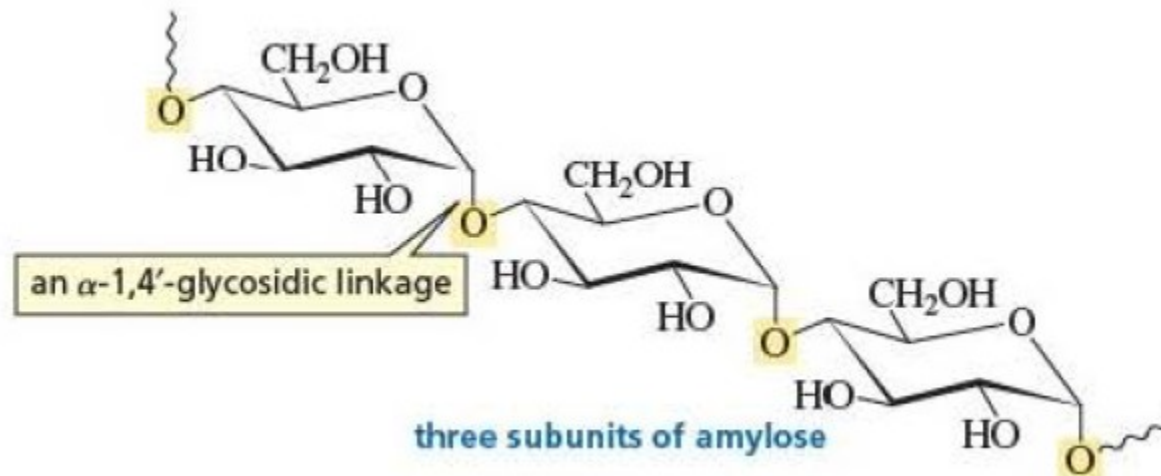
2.6.3. *Starch*

- **Starch**

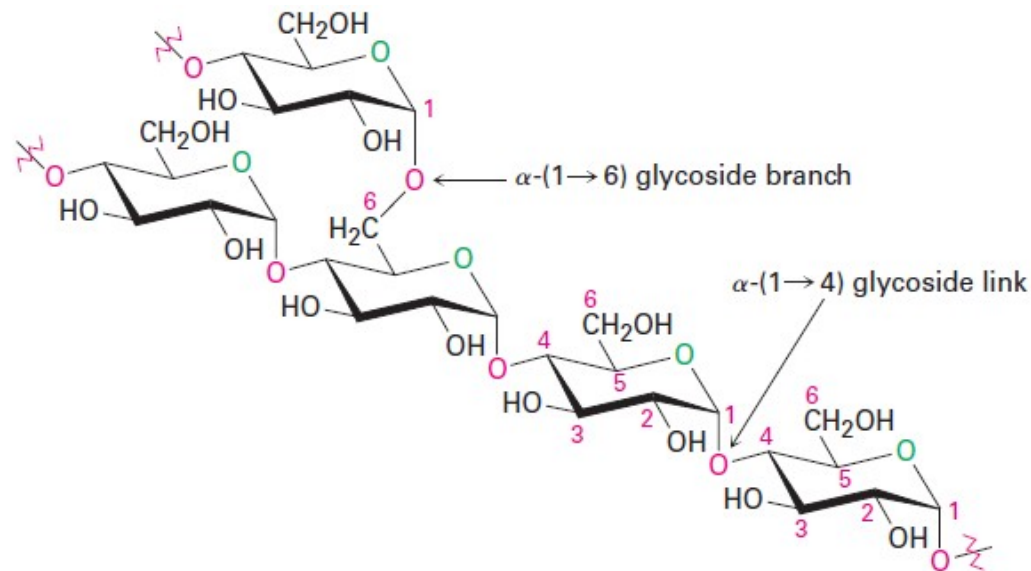
- Polymer of glucose

- **Amylose**

- **Unbrached polymer of D-glucose α -1,4-glycosidic linkages**



- Amylopectin is a branched polysaccharide
 - Main chain: **α -1,4-glycosidic linkages**
 - Branching : **α -1,6' glycosidic linkage**
 - **Limited solubility**



Amylopectin: α -(1 $\rightarrow$ 4) links
with α -(1 $\rightarrow$ 6) branches

2.6.4. Cellulose

- **Cellulose**

- β -1,4'-glycosidic linkages

- Intramolecular hydrogen bonds +
 - Intermolecular hydrogen bonds
 - Reduce water solubility +
 - Strong fibers

